



— LAUGHTALE · CCNA 200-301 V1.1 · PRINTABLE PACK

CCNA Printable *Study Pack*

The complete pack in one file — master plan, all 53 blueprint objectives, the 12-week calendar and five Packet Tracer labs.

53

Objectives

12

Weeks

5

Labs

13

High-yield

01 · MASTER PLAN

02 · BLUEPRINT

03 · CALENDAR

04 · LABS

■ Three ticks per objective: Studied · Anki · Proven

■ ★ HIGH YIELD = over-invest · STOP = hard depth limit

PRINT · A4 PORTRAIT · SINGLE COLUMN · BACKGROUND GRAPHICS ON

Study Master Plan

Scope source: Cisco CCNA Exam v1.1 (200-301) official blueprint, 2024 Cisco Systems. Exam format: 120 minutes. Multiple choice, multiple-select, drag-and-drop, and performance-based simulation/simlet items. No back button — an answered question is locked. Primary resource: Jeremy's IT Lab (Free CCNA v1.1 200-301 Complete Course — video + book + Anki). Secondary resource: Cisco Official Cert Guide (OCG), Volumes 1 & 2 — review pass only. Lab platform: Cisco Packet Tracer. This plan is split across four files — read/use them in this order:

1. 00_Index_and_How_To_Use.md — this file (notation, checkboxes, daily loop, rules)
2. 01_Blueprint_Breakdown.md — Section 1: all 53 sub-objectives, verb class, what to master, depth limits, high-yield ranking, do-not-study list
3. 02_Study_Calendar.md — Section 2: the 12-week week-by-week schedule
4. 03_Lab_Scenarios.md — Section 3: the five progressive Packet Tracer labs

How To Use This Document

Notation legend (used identically in every entry)

TAG	MEANING	TIME SPLIT
CONFIG	Blueprint verb is Configure and verify	70% hands-on CLI / 30% theory
CONCEPT	Blueprint verb is Describe / Explain / Compare / Define / Recognize	80% conceptual / 20% lab
INTERPRET	Blueprint verb is Interpret / Identify / Determine	50% reading real output / 50% theory
HY	High-yield. Disproportionate point share. Over-invest here.	—
STOP:	Depth constraint. Everything past this line is out of scope. Do not study it.	—

The three checkboxes under every objective

- ☐ Studied — video watched, book chapter read, notes written in your own words.
- ☐ Anki — flashcards unsuspended and in the daily review rotation.
- ☐ Proven — for **CONCEPT** / **INTERPRET**: explained out loud for 2 minutes with notes closed. For **CONFIG**: full command sequence typed on a blank CLI with no ? and no Tab completion.

An objective is not finished until all three are ticked. "Studied" alone is the most common reason people fail this exam.

Daily loop (non-negotiable, every study day)

- ☐ Anki reviews first — always. Never skip, never let the backlog build past one day.
- ☐ Five subnetting problems, timed. Target ≤45 seconds each by Week 4, ≤30 seconds by Week 10.
- ☐ New content block (video → book section → your own notes).
- ☐ Packet Tracer immediately after any **CONFIG** topic. Same session. Not "tomorrow."
- ☐ Two-minute verbal recap of what you just learned, out loud, notes closed.

Rules that carry the whole plan

1. The blueprint is the syllabus. If a topic is not in this document, Cisco is not testing it. Jeremy teaches some out-of-scope material (RIP, EIGRP, VTP, STP legacy timers in depth) because it aids understanding — watch those once, then let them go. They get zero exam points.
 2. Verb dictates method. You cannot pass a **CONFIG** objective by reading, and you cannot pass a **CONCEPT** objective by labbing. Match the method to the verb every time.
 3. Simulation items are worth multiple questions each. Getting a sim wrong costs far more than a multiple-choice miss. This is why **CONFIG** topics get 70% of your time despite being fewer in number.
 4. OCG is a second pass, not a first pass. Do not read it cover to cover. Use it to attack specific weak objectives identified by practice exams in Weeks 9–12.
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Overview

Part of the CCNA 200-301 v1.1 Study Master Plan — see 00_Index_and_How_To_Use.md for notation/checkbox conventions.
Next: 02_Study_Calendar.md.

Section 1 — Exam Blueprint Breakdown

Domain 1.0 — Network Fundamentals (20%)

Thirteen objectives. Two of them (1.6, 1.8) carry more weight than the other eleven combined, because subnetting math bleeds into every other domain on the exam.

1.1 Explain the role and function of network components

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 1

- Which OSI layer each device makes its forwarding decision at: router (L3/IP), L2 switch (L2/MAC), L3 switch (both, in hardware), NGFW (up to L7), IPS (inline, L7 inspection, blocks in real time), AP (802.11 ↔ 802.3 conversion), WLC (centralised control of lightweight APs via CAPWAP), endpoint, server.
- PoE standards and budgets: 802.3af (15.4 W), 802.3at / PoE+ (30 W), 802.3bt / UPoE and UPoE+ (60 W / 90 W). Know power is delivered over the same twisted pair as data, and typical draws (IP phone, AP, camera).
- IDS vs IPS: IDS is out-of-band and alerts; IPS is inline and drops.

STOP • DEPTH LIMIT

Do not learn specific Cisco model numbers, Firepower policy configuration, or IPS signature tuning. Do not configure PoE — there is no configure verb here.

☐ STUDIED ☐ ANKI ☐ PROVEN

1.2 Describe characteristics of network topology architectures

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 52 (LAN ARCHITECTURES), DAY 53 (WAN ARCHITECTURES)

- Two-tier (collapsed core): access + collapsed distribution/core. When it's the right answer — small to medium campus, cost-constrained.
- Three-tier: access (endpoint connectivity, PoE, port security), distribution (routing, VLAN boundary, policy, L2/L3 demarcation), core (high-speed transport, no policy, no filtering).
- Spine-leaf: every leaf connects to every spine, never leaf-to-leaf or spine-to-spine. Two hops maximum between any endpoints = predictable latency. Solves east-west traffic and removes STP-blocked-link waste. Data centre answer.

- WAN: leased line, MPLS L2VPN/L3VPN, Metro Ethernet, internet VPN, SD-WAN. Know what each is at a one-paragraph level.
- SOHO: single integrated box — router + switch + AP + firewall + NAT.
- On-premises vs cloud: IaaS / PaaS / SaaS responsibility boundaries, and the trade-offs (CapEx vs OpEx, control vs elasticity).

STOP • DEPTH LIMIT

No ACI, no VXLAN encapsulation detail, no MPLS label mechanics, no SD-WAN vEdge/vSmart component naming, no AWS/Azure service names.

☐ STUDIED ☐ ANKI ☐ PROVEN

1.3 Compare physical interface and cabling types

CLASS [CONCEPT]

YIELD LOW-MEDIUM

JEREMY DAY 2

- Single-mode fibre: small core (~9 µm), laser source, long distance (km range), more expensive. Multimode: larger core (50/62.5 µm), LED/VCSEL, shorter distance, cheaper.
- Copper twisted pair: Cat5e (1 Gbps), Cat6 (10 Gbps to 55 m), Cat6a (10 Gbps to 100 m). 100 m limit for all.
- Straight-through vs crossover and which device pairs need which (then know that MDIX auto makes it mostly moot on modern gear).
- Shared media (hub, half-duplex, one collision domain) vs point-to-point (switch, full-duplex, no collisions).

STOP • DEPTH LIMIT

Do not memorise every IEEE standard name (1000BASE-LX10 vs -EX vs -ZX distance tables). Know the categories and the reasoning, not the catalogue.

☐ STUDIED ☐ ANKI ☐ PROVEN

1.4 Identify interface and cable issues (collisions, errors, duplex/speed mismatch)

CLASS [INTERPRET]

YIELD MEDIUM

JEREMY DAY 9 (SWITCH INTERFACES)

- Read show interfaces output and name the fault from the counters. This is the actual exam skill.
- CRC errors → damaged cable, EMI, bad transceiver. CRC with no collisions often means duplex mismatch on the receiving side.
- Runt (<64 bytes) → collisions or duplex mismatch. Giants (>1518) → MTU/oversized frame issue.
- Late collisions → duplex mismatch (classic) or cable run exceeding spec.
- Input errors vs output errors vs drops — which direction is broken.
- Speed mismatch = link does not come up at all. Duplex mismatch = link comes up and performs terribly. Know this distinction cold; it's a favourite question.
- show interfaces status for a fast up/down/err-disabled and speed/duplex sweep.

STOP • DEPTH LIMIT

No optical power budget calculations, no TDR cable testing output, no SFP diagnostics.

☐ STUDIED ☐ ANKI ☐ PROVEN

1.5 Compare TCP to UDP

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 30

- TCP: connection-oriented, three-way handshake (SYN, SYN-ACK, ACK), four-way termination (FIN/ACK x2), sequence and acknowledgement numbers, sliding window / flow control, retransmission, 20-byte minimum header.

- UDP: connectionless, no handshake, no ordering, no retransmission, 8-byte header, four fields only (source port, dest port, length, checksum).
- Header fields for both, in order. You will get asked what a specific field does.
- Port numbers you must know by heart: FTP 20/21, SSH 22, Telnet 23, SMTP 25, DNS 53, DHCP 67/68, TFTP 69, HTTP 80, POP3 110, NTP 123, SNMP 161/162, HTTPS 443, Syslog 514, RADIUS 1812/1813, TACACS+ 49.
- Which applications pick which and why (VoIP/video/DNS/DHCP/TFTP → UDP; web/mail/file transfer/SSH → TCP).

STOP • DEPTH LIMIT

No congestion control algorithm detail (Reno, CUBIC, slow start maths). No TCP option fields beyond MSS awareness.

☐ STUDIED ☐ ANKI ☐ PROVEN

1.6 Configure and verify IPv4 addressing and subnetting ★ HIGH YIELD

CLASS [CONFIG]

YIELD HIGHEST IN THE EXAM

JEREMY DAYS 7, 8, 13, 14, 15

- Given any address and prefix, produce in ≤45 seconds: network ID, broadcast address, first usable host, last usable host, usable host count, dotted-decimal mask, and the next subnet's network ID.
- The reverse direction: given a host requirement, pick the correct prefix. Given a subnet count requirement, pick the correct prefix.
- VLSM: carve one block into differently-sized subnets in descending size order without overlap.
- Route summarisation: given a set of networks, produce the summary prefix.
- CLI: address a router interface, address an SVI, address a subinterface, no shutdown, and verify with show ip interface brief, show ip interface, show ip route, show running-config interface.
- Recognise up/up, up/down, administratively down/down and what each implies about layer 1 vs layer 2 vs configuration.

STOP • DEPTH LIMIT

Nothing. There is no rabbit hole here — this is the one objective where more time is always correct. It also underpins ACLs, OSPF wildcard masks, NAT pools, and DHCP scopes.

☐ STUDIED ☐ ANKI ☐ PROVEN

1.7 Describe private IPv4 addressing

CLASS [CONCEPT]

YIELD LOW (BUT ASSUMED EVERYWHERE)

JEREMY DAY 7

- RFC 1918: 10.0.0.0/8, 172.16.0.0/12 (172.16.0.0 – 172.31.255.255 — know the /12 boundary, it's a trap), 192.168.0.0/16.
- Not routable on the public internet; requires NAT to reach it.
- Adjacent special ranges you should recognise on sight: 127.0.0.0/8 loopback, 169.254.0.0/16 APIPA/link-local (what it means when a client has one — DHCP failed), 224.0.0.0/4 multicast.

STOP • DEPTH LIMIT

No RFC 6598 CGNAT (100.64.0.0/10) depth, no full IANA special-purpose registry.

☐ STUDIED ☐ ANKI ☐ PROVEN

1.8 Configure and verify IPv6 addressing and prefix ★ HIGH YIELD

CLASS [CONFIG]

YIELD HIGH

JEREMY DAYS 31, 32, 33

- Address compression and expansion rules, in both directions, flawlessly. Leading-zero removal, single :: only, and the "longest run of zeros" tie-break.
- IPv6 subnetting on nibble boundaries (/48 → /52 → /56 → /60 → /64). Know why /64 is the standard host prefix.

- CLI: enable IPv6 routing globally, assign a static global unicast address, assign an address using EUI-64, assign a link-local address manually, configure an IPv6 static route and default route.
- Verify with show ipv6 interface brief, show ipv6 interface, show ipv6 route, and IPv6 ping including pinging a link-local address (which requires specifying the exit interface).
- Understand that an interface has multiple IPv6 addresses simultaneously (link-local + one or more global) and which is used as source for what.

STOP • DEPTH LIMIT

No DHCPv6 server configuration, no SLAAC deep mechanics beyond RA/RS purpose, no OSPFv3, no IPv6 ACLs, no NAT64/DS-Lite/tunnelling.

☐ STUDIED ☐ ANKI ☐ PROVEN

1.9 Describe IPv6 address types

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAYS 31, 32

- Identify type from the leading bits on sight: GUA 2000::/3, ULA fc00::/7 (practically fd00::/8), link-local fe80::/10, multicast ff00::/8, unspecified ::, loopback ::1.
- Anycast: same address on multiple nodes, routed to the topologically nearest. Note there is no IPv6 broadcast — multicast replaces it.
- Well-known multicast groups: ff02::1 all nodes, ff02::2 all routers, ff02::5/ff02::6 OSPF, solicited-node ff02::1:ff00:0/104.
- Modified EUI-64, performed by hand: split the 48-bit MAC, insert FFFE in the middle, flip the 7th bit of the first byte (the U/L bit). Be able to do this in both directions.

STOP • DEPTH LIMIT

No multicast routing, no MLD, no Neighbour Discovery packet-by-packet breakdown beyond NS/NA/RS/RA purpose.

☐ STUDIED ☐ ANKI ☐ PROVEN

1.10 Verify IP parameters for Client OS (Windows, macOS, Linux)

CLASS [CONFIG] (VERIFY ONLY)

YIELD LOW-MEDIUM

JEREMY DAY 32 / LAB VIDEOS

- Windows: ipconfig /all (IP, mask, gateway, DNS, MAC, DHCP server), ping, tracert, nslookup, arp -a.
- Linux: ip addr, ip route, ip neigh, dig, legacy ifconfig.
- macOS: ifconfig, networksetup -getinfo <service>.
- The actual exam skill: read client output and spot the misconfiguration — wrong gateway (not in the client's own subnet), wrong mask (breaks local vs remote decisions), missing DNS, or a 169.254.x.x address meaning DHCP never answered.

STOP • DEPTH LIMIT

No OS-specific network stack tuning, no netsh scripting, no systemd-networkd or NetworkManager config files.

☐ STUDIED ☐ ANKI ☐ PROVEN

1.11 Describe wireless principles

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 55

- Non-overlapping channels: 2.4 GHz has only 1, 6, 11 at 20 MHz. 5 GHz has many (24+ depending on DFS/regulatory domain). Be able to explain co-channel interference vs adjacent-channel interference and which is worse.
- SSID vs BSSID vs ESS: SSID is the name, BSSID is the AP radio's MAC, ESS is multiple APs sharing an SSID.

- RF behaviour: attenuation, absorption, reflection, refraction, scattering, multipath; SNR and why it matters more than raw signal strength; the relationship between distance, data rate, and modulation.
- Encryption/authentication: WPA2-Personal (PSK) vs WPA2-Enterprise (802.1X/RADIUS); AES-CCMP; WPA3 with SAE and its improvement over the WPA2 four-way handshake.
- 2.4 GHz vs 5 GHz trade-off: range/penetration vs capacity/interference.

STOP • DEPTH LIMIT

No 802.11 frame-type breakdown, no antenna gain dBi maths, no RF site-survey methodology, no specific 802.11ax/be feature tables.

☐ STUDIED ☐ ANKI ☐ PROVEN

1.12 Explain virtualization fundamentals (server virtualization, containers, VRFs)

CLASS [CONCEPT]

YIELD LOW-MEDIUM

JEREMY DAY 54 (PARTS 1-3)

- Type 1 hypervisor (bare metal — ESXi, Hyper-V, KVM) vs Type 2 (hosted — VirtualBox, VMware Workstation). Which is used in a data centre and why.
- Virtual switch role: how VMs reach the physical network, and why the physical uplink is typically a trunk.
- Containers vs VMs: containers share the host kernel, no guest OS, seconds to start, megabytes not gigabytes. VMs give stronger isolation.
- VRF: multiple independent routing tables on one device. The one-line framing that wins questions — "VRFs are to Layer 3 what VLANs are to Layer 2."

STOP • DEPTH LIMIT

No vSphere/ESXi administration, no Docker commands, no Kubernetes at all, no VRF-lite configuration (there is no configure verb).

☐ STUDIED ☐ ANKI ☐ PROVEN

1.13 Describe switching concepts

CLASS [CONCEPT]

YIELD MEDIUM-HIGH

JEREMY DAYS 5, 6

- MAC learning: the switch reads the source MAC of an incoming frame and maps it to the ingress port and VLAN. Aging default 300 seconds on Cisco.
- Frame switching: destination MAC found in the CAM table → forward out that one port only.
- Frame flooding: destination is broadcast, unknown unicast, or multicast → out every port in the VLAN except the ingress port.
- MAC address table / CAM table structure: MAC ↔ port ↔ VLAN ↔ type (dynamic/static).
- The exam skill: given a topology and a specific sequence of frames, state the exact contents of each switch's MAC table afterwards, and state for each frame whether it was switched or flooded. Practise this on paper — it is a guaranteed question type.
- show mac address-table, and clear mac address-table dynamic.

STOP • DEPTH LIMIT

No store-and-forward vs cut-through vs fragment-free switching detail (removed from current scope), no TCAM/ASIC architecture.

☐ STUDIED ☐ ANKI ☐ PROVEN

Domain 2.0 — Network Access (20%)

The heaviest simulation domain. VLANs, trunking, EtherChannel, and STP interpretation are where performance-based items concentrate. Note that in v1.1, objective 2.9 was softened from configure to interpret — the wireless GUI is now a reading exercise, not a building one.

2.1 Configure and verify VLANs (normal range) spanning multiple switches ★ HIGH YIELD

CLASS [CONFIG] YIELD VERY HIGH JEREMY DAYS 16, 17, 18

- Create a VLAN, name it, assign an access port, verify with show vlan brief.
- Data vs voice access ports on the same physical interface: a voice VLAN carries tagged voice traffic while the data VLAN stays untagged on the same port.
- Default VLAN 1 behaviour, why it's poor practice to use it, and normal range (1–1005) vs extended range (1006–4094 — awareness only).
- VLANs spanning multiple switches: the same VLAN ID must exist on every switch in the path and be allowed on the trunks between them. A missing VLAN on a transit switch is the single most common lab failure — expect to troubleshoot exactly this.
- InterVLAN connectivity: router-on-a-stick (subinterfaces with encapsulation dot1q) and L3 switch SVIs (ip routing + SVI per VLAN). Know when each is appropriate.
- Verify with show vlan brief, show interfaces switchport, show interfaces trunk, show ip route.

STOP • DEPTH LIMIT

No VTP configuration (Jeremy covers it in Day 19 for context — watch once, do not lab it; it is not on the v1.1 blueprint). No private VLANs. No extended-range VLAN configuration.

☐ STUDIED ☐ ANKI ☐ PROVEN

2.2 Configure and verify interswitch connectivity ★ HIGH YIELD

CLASS [CONFIG] YIELD VERY HIGH JEREMY DAYS 17, 18

- Statically configure a trunk port (mode trunk, and where required set the encapsulation first on older platforms).
- 802.1Q tag structure: 4 bytes inserted after the source MAC, 12-bit VLAN ID field, TPID 0x8100. Native VLAN frames leave untagged.
- Native VLAN: change it, and understand the security reason (VLAN hopping / double-tagging) and the operational reason (a native VLAN mismatch generates errors and merges two VLANs' traffic).
- Prune the allowed VLAN list on a trunk, and add/remove from it correctly without wiping the list.
- Troubleshoot a trunk that won't form: mode mismatch, native VLAN mismatch, VLAN not allowed, VLAN not defined.
- Verify with show interfaces trunk (the single most useful switching command on the exam) and show interfaces switchport.

STOP • DEPTH LIMIT

DTP exists and you should know what dynamic auto/dynamic desirable do and why hardcoding is better practice — but do not build a lab around DTP negotiation states. No ISL (dead).

☐ STUDIED ☐ ANKI ☐ PROVEN

2.3 Configure and verify Layer 2 discovery protocols (CDP and LLDP)

- Enable/disable CDP globally and per-interface. Same for LLDP — and note LLDP is disabled by default on Cisco while CDP is enabled by default.
- LLDP's transmit and receive are configured separately per interface. This is a favourite exam detail.
- Default timers: CDP 60 s hello / 180 s hold. LLDP 30 s / 120 s.
- Read show cdp neighbors and show cdp neighbors detail (the detail version gives the neighbour's IP address — that's why it matters). Same for show lldp neighbors / detail.
- Use the output to reconstruct a topology diagram. That is the exam task: "given this CDP output, which device is connected to which port."
- Security angle: CDP leaks device model and IOS version, so disable it on untrusted-facing ports.

STOP • DEPTH LIMIT

No CDP/LLDP TLV field enumeration.

☐ STUDIED ☐ ANKI ☐ PROVEN

2.4 Configure and verify (Layer 2/Layer 3) EtherChannel (LACP) ★ HIGH YIELD

- LACP modes: active and passive. Active/active and active/passive form a channel; passive/passive does not. PAGP (desirable/auto) and static on exist — know the mode-combination matrix.
- Layer 2 EtherChannel: bundle physical ports into a port-channel, then configure trunking/VLAN settings on the port-channel interface, not the members.
- Layer 3 EtherChannel: no switchport on members, create the port-channel, apply the IP address to the port-channel interface.
- The consistency requirement: every member port must match on speed, duplex, mode (access/trunk), native VLAN, and allowed VLAN list. Mismatch = port excluded from the bundle. This is the standard troubleshooting scenario.
- Load-balancing methods (src-mac, dst-mac, src-dst-ip etc.) — know that the hash determines which physical link a flow uses, and that a single flow never spans links.
- Verify with show etherchannel summary (read the flags — P bundled, D down, s suspended, I stand-alone), show etherchannel port-channel, show interfaces port-channel.

STOP • DEPTH LIMIT

No advanced hashing algorithm internals, no cross-stack/MLAG/vPC.

☐ STUDIED ☐ ANKI ☐ PROVEN

2.5 Interpret basic operations of Rapid PVST+ Spanning Tree Protocol ★ HIGH YIELD

- Root bridge election: lowest bridge ID wins = priority (default 32768) + extended system ID (the VLAN number) + MAC address. Compute this by hand from a diagram. Priority must be set in increments of 4096.
- Root port selection order: lowest root path cost → lowest sender bridge ID → lowest sender port ID → lowest local port ID. Know the tie-break order in sequence.
- Port cost defaults for RSTP/short mode: 10 Mbps = 100, 100 Mbps = 19, 1 Gbps = 4, 10 Gbps = 2.
- Port roles: root, designated, alternate, backup, disabled. Port states in RSTP: discarding, learning, forwarding (RSTP collapses blocking/listening into discarding — be able to map the legacy STP states to the RSTP ones).
- Link types: point-to-point (full duplex), shared (half duplex), edge (PortFast).
- PortFast: skip straight to forwarding on an access port with an endpoint. Never on a switch-to-switch link.
- Protection features and what each stops: BPDU guard (err-disables a PortFast port that receives a BPDU), BPDU filter (suppresses BPDUs), root guard (blocks a superior BPDU from taking over as root, puts port in root-inconsistent), loop guard (protects against unidirectional link failure causing a blocked port to wrongly go forwarding).
- Set primary and secondary root for a VLAN and know what that command actually does to the priority value.

- Verify with show spanning-tree, show spanning-tree vlan <id>, show spanning-tree root, show spanning-tree interface <int> detail.
- The exam skill: given a diagram with bridge priorities and MACs, mark every port's role and state. Do this by hand, repeatedly.

STOP • DEPTH LIMIT

The verb is interpret, not configure. You still need the root-priority and PortFast/BPDU-guard commands, but do not study MST, do not study PVST+ legacy timer tuning (hello/forward delay/max age maths), and do not study STP convergence timer optimisation.

☐ STUDIED ☐ ANKI ☐ PROVEN

2.6 Describe Cisco Wireless Architectures and AP modes

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 56

- Autonomous AP (standalone, self-configured) vs lightweight AP + WLC (split-MAC, centralised).
- Split-MAC: which functions stay on the AP (real-time — beacons, ACKs, encryption, buffering) and which move to the WLC (management — association, authentication, RRM, security policy).
- CAPWAP: two tunnels, control (UDP 5246) and data (UDP 5247). Tunnels the traffic from AP to WLC.
- WLC deployment models: unified/on-prem appliance, embedded, cloud-based, Mobility Express, and the concept of a centralised vs FlexConnect data path.
- AP modes: local (default), FlexConnect, monitor, sniffer, rogue detector, SE-Connect, bridge/mesh. Know which one you'd pick for a given requirement (e.g. a branch site with a slow WAN link → FlexConnect).

STOP • DEPTH LIMIT

No WLC HA/SSO configuration, no RRM algorithm internals, no mesh RAP/MAP design.

☐ STUDIED ☐ ANKI ☐ PROVEN

2.7 Describe physical infrastructure connections of WLAN components

CLASS [CONCEPT]

YIELD LOW-MEDIUM

JEREMY DAY 56

- AP switchport: access port if one SSID/VLAN and traffic is tunnelled to the WLC in local mode; trunk port when the AP is autonomous or in FlexConnect with local switching of multiple VLANs. Be able to justify which and why.
- WLC ports: distribution system port, service port, redundancy port, console. WLC interfaces: management, virtual, dynamic (per-WLAN), AP-manager.
- LAG on the WLC: bundles distribution ports into one logical link; know that it's typically all-or-nothing and requires a matching EtherChannel on the switch side.
- PoE feeding the AP from the access switch, and the power-budget consequence.

STOP • DEPTH LIMIT

No WLC port-by-port cabling specifications per hardware model.

☐ STUDIED ☐ ANKI ☐ PROVEN

2.8 Describe network device management access

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 4, DAY 42, DAY 48

- Console (out-of-band, physical, works with no network), Telnet (TCP 23, cleartext — never use), SSH (TCP 22, encrypted — always use), HTTP (80) vs HTTPS (443) for GUI access.

- TACACS+ vs RADIUS: TACACS+ is Cisco-originated, TCP 49, encrypts the entire payload, and separates authentication/authorisation/accounting — so it's the device-administration choice. RADIUS is UDP 1812/1813, encrypts only the password, and combines authentication and authorisation — so it's the network-access/802.1X choice.
- Cloud-managed (Meraki-style): device phones home to a cloud controller; separation of management plane from data plane.
- In-band vs out-of-band management, and why OOB matters when the network is broken.

STOP • DEPTH LIMIT

No ISE policy configuration, no TACACS+ packet format.

☐ STUDIED ☐ ANKI ☐ PROVEN

2.9 Interpret the wireless LAN GUI configuration for client connectivity

CLASS [INTERPRET]

YIELD MEDIUM

JEREMY DAY 58

- Walk the WLC GUI WLAN creation flow and be able to read a screenshot and say what it will do: WLAN ID/SSID/profile name, the interface (which maps the WLAN to a VLAN), status enable/disable, broadcast SSID on/off.
- Security tab: Layer 2 security choice (WPA2/WPA3, PSK vs 802.1X), AKM, encryption cipher, the PSK field, and where a RADIUS server gets attached for Enterprise.
- QoS tab: Platinum (voice) / Gold (video) / Silver (best effort, default) / Bronze (background). Know the ordering.
- Advanced tab: session timeout, client exclusion, DHCP required, FlexConnect local switching, band select, client load balancing.

STOP • DEPTH LIMIT

The v1.1 verb is interpret, not configure. You are expected to read and reason about GUI screens, not build them from memory. Do not memorise menu paths for every WLC firmware version. Objective 5.10 is the one place a GUI configure verb still applies (WPA2 PSK).

☐ STUDIED ☐ ANKI ☐ PROVEN

Domain 3.0 — IP Connectivity (25%)

The largest single domain. Only five objectives, but three of them (3.1, 3.3, 3.4) are dense and heavily simulated. OSPFv2 is the highest-value single technology on the exam.

3.1 Interpret the components of routing table ★ HIGH YIELD

CLASS [INTERPRET] YIELD VERY HIGH JEREMY DAY 11 (ROUTING FUNDAMENTALS)

- Read any line of show ip route and name every field. Given O 192.168.10.0/24 [110/65] via 10.0.0.2, 00:04:11, GigabitEthernet0/1, you must instantly identify: source code, prefix, mask, administrative distance (110), metric (65), next hop, uptime, exit interface.
- Routing protocol codes: C connected, L local, S static, S* static default, O OSPF, O IA OSPF inter-area, D EIGRP, R RIP, B BGP.
- Default administrative distances, memorised: connected 0, static 1, eBGP 20, EIGRP 90, OSPF 110, RIP 120, external EIGRP 170, iBGP 200, unreachable 255.
- Gateway of last resort: what sets it, how it displays, and what happens when it isn't set.
- The difference between a C connected route and its paired L local /32 host route.
- show ip route <prefix> for the detailed single-route view.

STOP • DEPTH LIMIT

No BGP path-attribute detail, no EIGRP metric formula (K-values), no route redistribution.

☐ STUDIED ☐ ANKI ☐ PROVEN

3.2 Determine how a router makes a forwarding decision by default ★ HIGH YIELD

CLASS [INTERPRET] YIELD HIGH JEREMY DAYS 11, 12 (LIFE OF A PACKET)

- The decision order, in this exact sequence: 1. Longest prefix match (most specific route wins, full stop — AD and metric never override a longer match). 2. Administrative distance (used only to choose between routes to the same prefix learned from different sources). 3. Metric (used only to choose between routes to the same prefix from the same protocol).
- Given a routing table and a destination IP, state the exact chosen route. This appears in some form on almost every attempt.
- Equal-cost multi-path: same prefix, same AD, same metric → both installed, load balanced.
- Life of a packet end to end: ARP, MAC rewrite at every hop, TTL decrement, source/destination IP never changing (absent NAT). Being able to narrate this is worth a surprising number of points.

STOP • DEPTH LIMIT

No CEF/FIB/adjacency-table internals, no per-packet vs per-destination load-balancing platform behaviour.

☐ STUDIED ☐ ANKI ☐ PROVEN

3.3 Configure and verify IPv4 and IPv6 static routing ★ HIGH YIELD

CLASS [CONFIG] YIELD VERY HIGH JEREMY DAY 11 (STATIC ROUTING), DAY 33

- Network route (a subnet), host route (/32 or /128), default route (0.0.0.0/0 and ::/0).
- Three ways to specify a static route — next-hop IP only, exit interface only, or both. Know the recursive-lookup cost of next-hop-only, and why exit-interface-only on a multi-access (Ethernet) segment causes ARP problems. Fully-specified is safest on Ethernet.

- Floating static: raise the AD above the dynamic protocol's so it only installs when the primary route dies. This is the standard backup-link question.
- IPv6 equivalents including a static route via a link-local next hop (which requires the exit interface to be specified).
- Verify: show ip route static, show ipv6 route static, and prove it by shutting the primary link and watching the floating static install.

STOP • DEPTH LIMIT

No policy-based routing, no VRF-aware static routes, no track/IP SLA object tracking.

☐ STUDIED ☐ ANKI ☐ PROVEN

3.4 Configure and verify single area OSPFv2 ★ HIGH YIELD

CLASS [CONFIG]

YIELD HIGHEST SINGLE TECHNOLOGY ON THE EXAM

JEREMY DAYS 24, 26, 27, 28

- Enable OSPF, set a process ID (locally significant — know that), and advertise networks with a wildcard mask. Also know the per-interface method (ip ospf <pid> area <area>) as an alternative.
- Router ID selection order: manually configured RID → highest loopback IP → highest active physical interface IP. Know that changing it requires a process clear/reload to take effect.
- Neighbour states in order: Down → Init → 2-Way → ExStart → Exchange → Loading → Full. Know what happens at each and where a stuck adjacency points (stuck in 2-Way on a broadcast segment is normal between two DROTHERs; stuck in ExStart usually means MTU mismatch).
- Requirements for adjacency, all of which must match: area ID, hello and dead intervals, subnet/mask on the segment, authentication, MTU, and the interfaces must not be passive. Given a broken lab, checking these five in order is the whole troubleshooting method.
- Network types: broadcast (Ethernet — elects DR/BDR, hello 10 / dead 40) vs point-to-point (no DR/BDR, hello 10 / dead 40). Know how to change the interface network type and why you'd force point-to-point on an Ethernet link between exactly two routers.
- DR/BDR election: highest OSPF interface priority wins; tie broken by highest router ID. Priority 0 means never DR/BDR. The election is non-preemptive — a new higher-priority router does not take over until the current DR fails. This non-preemption point is a guaranteed question.
- Passive interfaces: stop hellos on a LAN with no OSPF neighbours while still advertising the subnet. Know passive-interface default and the per-interface exception form.
- Cost/metric: cost = reference bandwidth (100 Mbps default) ÷ interface bandwidth. Know why the default reference bandwidth is a problem on gigabit links and how to change it. Also know how to set cost manually per interface.
- Default route injection with default-information originate.
- Verify: show ip ospf neighbor (states, DR/BDR roles), show ip ospf interface brief, show ip ospf interface <int> (network type, cost, timers, priority), show ip protocols, show ip route ospf.

STOP • DEPTH LIMIT

Hard stop at single area. Do not study: multi-area design, ABR/ASBR behaviour, LSA types 1–7, virtual links, stub/NSSA/ totally-stubby areas, OSPFv3, OSPF authentication configuration, or route summarisation between areas. Jeremy covers some of this for completeness — watch, don't lab. Also: RIP and EIGRP (Jeremy Day 25) are not on the v1.1 blueprint at all. Watch once for AD context, then move on.

☐ STUDIED ☐ ANKI ☐ PROVEN

3.5 Describe the purpose, functions, and concepts of first hop redundancy protocols

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 29

- The problem FHRPs solve: an endpoint has one default gateway, so a redundant router is useless without a shared virtual IP.
- Virtual IP and virtual MAC concept — the hosts point at the VIP and never know a failover happened.

- HSRP (Cisco proprietary, active/standby, virtual MAC 0000.0C07.ACxx for v1, priority 100 default, preemption off by default), VRRP (open standard, master/backup, 0000.5E00.01xx), GLBP (Cisco, provides actual load balancing across multiple forwarders, not just redundancy).
- Load-balancing across FHRP groups by making different routers active for different VLANs.
- Note: v1.1 makes this a describe objective, but Lab 4 in this plan still has you configure HSRP — because building it is the fastest way to actually understand it, and it's a real-job skill.

STOP • DEPTH LIMIT

Describe verb only. No HSRP authentication, no interface tracking decrement tuning depth, no GLBP configuration.

☐ STUDIED ☐ ANKI ☐ PROVEN

Domain 4.0 — IP Services (10%)

Nine objectives for 10% of the exam — so each one is shallow. Resist depth here. Only 4.1, 4.2, 4.6, and 4.8 carry a configure verb.

4.1 Configure and verify inside source NAT using static and pools

CLASS [CONFIG]

YIELD MEDIUM-HIGH

JEREMY DAYS 44, 45

- The four address types and what each means: inside local, inside global, outside local, outside global. Fill in a NAT translation table from a scenario — this is the classic question.
- Mark inside and outside interfaces correctly (ip nat inside / ip nat outside). Forgetting this is the number one reason NAT "doesn't work" in a lab.
- Static NAT: a permanent one-to-one mapping.
- Dynamic NAT with a pool: define the pool, define an ACL matching the inside source addresses, bind them.
- PAT: overload a pool or overload the outside interface address; understand that port numbers distinguish the sessions.
- Verify with show ip nat translations and show ip nat statistics; clear ip nat translation * to reset.

STOP • DEPTH LIMIT

No NAT64, no CGNAT, no NAT virtual interfaces, no port forwarding beyond basic static NAT with port.

☐ STUDIED ☐ ANKI ☐ PROVEN

4.2 Configure and verify NTP operating in client and server mode

CLASS [CONFIG]

YIELD LOW-MEDIUM

JEREMY DAY 37

- Point a device at an NTP server; configure a device as an NTP master with a stratum.
- Stratum concept: stratum 0 reference clock, each hop adds one, stratum 16 = unsynchronised.
- Why it matters operationally: log correlation, certificate validity, Kerberos.
- Verify with show ntp status (synchronised/unsynchronised, stratum) and show ntp associations (the * marks the chosen peer).
- Set the timezone and know clock set for manual time.

STOP • DEPTH LIMIT

No NTP authentication, no symmetric active/passive or broadcast modes in depth.

☐ STUDIED ☐ ANKI ☐ PROVEN

4.3 Explain the role of DHCP and DNS within the network

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAYS 38, 39

- DHCP DORA sequence — Discover (broadcast), Offer, Request (broadcast), Acknowledge. Know which are broadcast and which are unicast, and the ports (67 server / 68 client).
- What a DHCP scope hands out: address, mask, default gateway, DNS servers, lease time, domain name.
- DNS: resolving a name to an address, recursive vs authoritative at a conceptual level, A vs AAAA vs CNAME vs MX vs PTR records, UDP 53 (TCP 53 for zone transfers/large responses).
- The troubleshooting logic: "can ping by IP but not by name" → DNS. "Client has 169.254.x.x" → DHCP unreachable.

STOP · DEPTH LIMIT

No DNSSEC, no zone file syntax, no BIND administration.

☐ STUDIED ☐ ANKI ☐ PROVEN

4.4 Explain the function of SNMP in network operations

CLASS [CONCEPT]

YIELD LOW

JEREMY DAY 40

- Manager / agent / MIB / OID relationship.
- Get, GetNext, GetBulk, Set, Trap, Inform. Know that a trap is unacknowledged and an inform is acknowledged — that single distinction is the most-asked SNMP fact.
- Versions: v1 and v2c use cleartext community strings; v3 adds authentication, integrity, and encryption. v3 is the answer to any security-flavoured SNMP question.
- Ports 161 (polling) and 162 (traps/informs).

STOP · DEPTH LIMIT

No SNMP configuration (no configure verb), no MIB tree walking, no OID memorisation.

☐ STUDIED ☐ ANKI ☐ PROVEN

4.5 Describe the use of syslog features, including facilities and severity levels

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 41

- Severity levels 0–7, memorised with a mnemonic: 0 Emergency, 1 Alert, 2 Critical, 3 Error, 4 Warning, 5 Notification, 6 Informational, 7 Debugging. ("Every Awesome Cisco Engineer Will Need Ice cream Daily.")
- Lower number = more severe. Configuring a logging level of 4 captures levels 0 through 4 — this inclusive behaviour is the exam trap.
- Message anatomy: %FACILITY-SEVERITY-MNEMONIC: description. Be able to read %LINK-3-UPDOWN and state the severity.
- Logging destinations: console, monitor (VTY), buffered, and a remote syslog server (UDP 514).
- Facility concept at a naming level only.

STOP · DEPTH LIMIT

No syslog server software administration, no RFC 5424 structured-data format.

☐ STUDIED ☐ ANKI ☐ PROVEN

4.6 Configure and verify DHCP client and relay

CLASS [CONFIG]

YIELD MEDIUM

JEREMY DAY 39

- Configure a router interface as a DHCP client (obtain its address via DHCP).
- Configure a DHCP relay agent (ip helper-address) on the SVI/subinterface facing the clients, pointing at the server. Understand precisely why it's needed: DHCP Discover is a broadcast and routers don't forward broadcasts. The relay converts it to a unicast and inserts the client subnet so the server picks the right scope.
- Configure a Cisco router as a DHCP server: pool, network, default-router, dns-server, and excluded addresses. Know that you exclude before defining the pool in practice.
- Verify with show ip dhcp binding, show ip dhcp pool, show ip dhcp conflict, and confirm from the client side.

STOP · DEPTH LIMIT

No DHCP option 82 configuration depth (it belongs to 5.7 DHCP snooping awareness), no DHCPv6.

☐ STUDIED ☐ ANKI ☐ PROVEN

4.7 Explain the forwarding per-hop behavior (PHB) for QoS

CLASS [CONCEPT] YIELD LOW-MEDIUM JEREMY DAYS 46, 47

- Classification (identify the traffic) and marking (set a value so downstream devices don't have to re-classify). CoS is the Layer 2 marking in the 802.1Q tag (3 bits, 0–7); DSCP is the Layer 3 marking in the IP ToS byte (6 bits).
- Key DSCP values: EF (Expedited Forwarding, DSCP 46) for voice; AF classes; CS classes; default 0.
- Queuing and congestion management: what a scheduler does, priority/LLQ for voice.
- Congestion avoidance: tail drop vs WRED.
- Policing vs shaping — policing drops or re-marks excess traffic immediately, shaping buffers it and sends it later. Policing is bursty, shaping is smooth. This comparison is the most likely QoS question.
- Trust boundary: why you trust an IP phone's markings but not a PC's.

STOP • DEPTH LIMIT

Explain verb only. No MQC configuration (class-map/policy-map/service-policy), no queue-depth tuning, no bandwidth-reservation maths.

☐ STUDIED ☐ ANKI ☐ PROVEN

4.8 Configure network devices for remote access using SSH ★ HIGH YIELD

CLASS [CONFIG] YIELD HIGH JEREMY DAY 42

- The full, ordered sequence — you will be asked to produce it and a missing step breaks it: set a hostname, set a domain name, generate an RSA key pair (1024 bits or more for SSHv2), create a local user with a privilege level and secret, configure the VTY lines for local login and to accept SSH transport only, and force version 2.
- Why the hostname and domain name must be set first (they seed the key's fully qualified name).
- VTY line hardening: exec-timeout, login local, transport input ssh, and optionally an ACL applied with access-class.
- enable secret vs enable password, and service password-encryption (weak, type 7, reversible — know that it is not real security).
- Verify with show ip ssh, show ssh, show users, and by actually SSHing in from another device.

STOP • DEPTH LIMIT

No SSH public-key authentication, no certificate-based auth, no AAA server integration configuration.

☐ STUDIED ☐ ANKI ☐ PROVEN

4.9 Describe the capabilities and functions of TFTP/FTP in the network

CLASS [CONCEPT] YIELD LOW JEREMY DAY 43

- TFTP: UDP 69, no authentication, no directory listing, tiny and simple. Used for IOS image transfer, config backup, and PXE boot.
- FTP: TCP 20 (data) and 21 (control), authenticated, supports directory operations, active vs passive mode at a conceptual level.
- The network-engineering use case: backing up and restoring running/startup configs and IOS images to/from a server.

STOP • DEPTH LIMIT

Lowest-yield objective in the blueprint. One Anki pass, then leave it. No SFTP/FTPS/SCP depth, no copy command drilling beyond basic awareness.

☐ STUDIED ☐ ANKI ☐ PROVEN

Domain 5.0 — Security Fundamentals (15%)

Mostly conceptual, with three sharp exceptions: ACLs (5.6) and Layer 2 security (5.7) are heavily simulated, and 5.10 is the one remaining GUI configure task.

5.1 Define key security concepts (threats, vulnerabilities, exploits, mitigation)

CLASS [CONCEPT]

YIELD LOW-MEDIUM

JEREMY DAY 48

- The four definitions and how they relate: a vulnerability is the weakness, a threat is the potential danger, an exploit is the mechanism used against the vulnerability, and mitigation is the countermeasure. Getting these four straight answers most questions here.
- CIA triad: confidentiality, integrity, availability — and which one a given attack breaks.
- Attack vocabulary: DoS/DDoS, reflection/amplification, spoofing (MAC, IP, DHCP, ARP), on-path/MITM, reconnaissance, social engineering (phishing, spear phishing, whaling, vishing, smishing), tailgating, password attacks (brute force, dictionary, credential stuffing), malware types.

STOP · DEPTH LIMIT

No threat-intelligence frameworks, no MITRE ATT&CK, no incident-response process models.

☐ STUDIED ☐ ANKI ☐ PROVEN

5.2 Describe security program elements (user awareness, training, physical access control)

CLASS [CONCEPT]

YIELD LOW

JEREMY DAY 48

- User awareness (broad, passive — posters, simulated phishing) vs training (formal, structured, role-specific) vs physical access control (locks, badges, mantraps, cameras, secured racks).
- Distinguish these three cleanly. The questions here are almost always "which category does this example belong to."

STOP · DEPTH LIMIT

One Anki pass. No compliance framework study (ISO 27001, NIST CSF).

☐ STUDIED ☐ ANKI ☐ PROVEN

5.3 Configure and verify device access control using local passwords

CLASS [CONFIG]

YIELD MEDIUM

JEREMY DAYS 4, 48

- Console line password, VTY line password, enable secret vs enable password, local username/secret pairs with privilege levels.
- service password-encryption (type 7, reversible, cosmetic) vs secret (type 5 MD5 / type 8/9 scrypt — hashed, not reversible). Know which is actually secure.
- login vs login local and what each implies about where credentials come from.
- exec-timeout and logging synchronous on lines.
- Verify with show running-config | section line, and show privilege / show users.

STOP · DEPTH LIMIT

No AAA configuration (aaa new-model and TACACS+/RADIUS setup is beyond the configure scope — 2.8 and 5.8 only require describing/comparing them).

☐ STUDIED ☐ ANKI ☐ PROVEN

5.4 Describe security password policy elements

CLASS [CONCEPT]

YIELD LOW

JEREMY DAY 48

- Complexity, length, expiration/rotation, reuse history, lockout thresholds, secure storage.
- Multifactor authentication as the headline alternative — the three factors: something you know, something you have, something you are. Be able to identify whether a given pair is genuinely multifactor (two passwords is not).
- Certificates and biometrics as password alternatives, and the trade-offs of each (biometrics can't be changed once compromised).
- Password managers and passphrases.

STOP · DEPTH LIMIT

No PKI hierarchy depth, no certificate enrolment process, no biometric FAR/FRR maths.

☐ STUDIED ☐ ANKI ☐ PROVEN

5.5 Describe IPsec remote access and site-to-site VPNs

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY V1.1 VPN VIDEO (WAN ARCHITECTURES / DAY 53 AREA)

- Site-to-site: gateway-to-gateway tunnel, always on, transparent to hosts, connects two whole networks. Remote-access: client-to-gateway, on-demand, per-user, typically SSL/TLS-based in modern deployments.
- What IPsec provides: confidentiality (encryption), integrity (hashing), authentication (pre-shared key or certificates), anti-replay.
- The two protocols: AH (authentication and integrity, no encryption) and ESP (encryption and integrity). ESP is what's actually used. That AH doesn't encrypt is the classic question.
- Tunnel mode (encrypts the whole original packet and adds a new IP header — used site-to-site) vs transport mode (encrypts payload only).
- IKE's role in negotiating the security association, at a one-paragraph level. GRE over IPsec: why GRE is added (it carries multicast/routing protocols, which plain IPsec can't).

STOP · DEPTH LIMIT

Describe verb only. No crypto map or IKEv2 proposal configuration. No DMVPN, no FlexVPN, no cipher-suite tables.

☐ STUDIED ☐ ANKI ☐ PROVEN

5.6 Configure and verify access control lists ★ HIGH YIELD

CLASS [CONFIG]

YIELD VERY HIGH

JEREMY DAYS 34, 35

- Wildcard masks, fluently and in both directions. This is subnetting again, inverted. 0 = must match, 1 = don't care. Know the shortcuts: host = 0.0.0.0, any = 255.255.255.255.
- Standard ACLs (1–99, 1300–1999): match source only. Placement rule: as close to the destination as possible, because they'd otherwise block too much.
- Extended ACLs (100–199, 2000–2699): match source, destination, protocol, and port. Placement rule: as close to the source as possible, to drop unwanted traffic early.
- Named ACLs (standard and extended) — and the key operational advantage: sequence numbers let you insert or delete an individual line without rebuilding the list. Know how to do that.

- Implicit deny any at the end of every ACL. Every ACL needs at least one permit or it blocks everything. This causes more lab failures than anything else.
- Top-down first-match processing: order matters, and a more specific rule must come before a broader one.
- Applying an ACL: to an interface with a direction (in or out), and to VTY lines with access-class.
- Verify with show access-lists (read the match counters — they prove traffic is hitting the rules), show ip interface <int> (shows which ACL is applied where), and show running-config.

STOP • DEPTH LIMIT

No IPv6 ACLs, no time-based ACLs, no reflexive or dynamic (lock-and-key) ACLs, no object groups.

☐ STUDIED ☐ ANKI ☐ PROVEN

5.7 Configure and verify Layer 2 security features (DHCP snooping, DAI, port security) ★ HIGH YIELD

CLASS [CONFIG]

YIELD HIGH

JEREMY DAYS 49, 50, 51

- Port security: enable it (requires the port be statically access or trunk first — it won't apply to a dynamic port), set maximum MAC count, choose the learning method (static, dynamic, or sticky), and set the violation mode.
- The three violation modes and their exact differences: protect (drops offending traffic silently, no log, no counter), restrict (drops, logs, increments the violation counter), shutdown (err-disables the port — this is the default). Know how to recover an err-disabled port (shut/no shut, or errdisable recovery).
- Aging and sticky MAC persistence (sticky entries go into the running-config, so they're lost on reload unless saved).
- DHCP snooping: the point is to stop a rogue DHCP server. Enable globally and per-VLAN, then designate trusted ports (uplinks toward the legitimate server and other switches); everything else is untrusted by default and has its DHCP server messages dropped. Know about rate limiting and Option 82 insertion, and the binding table it builds (MAC ↔ IP ↔ VLAN ↔ port ↔ lease).
- Dynamic ARP Inspection: stops ARP spoofing / on-path attacks. Enable per-VLAN, set trusted ports, and understand that DAI depends on the DHCP snooping binding table — which is why snooping must be configured first. Static ARP ACLs cover non-DHCP hosts.
- Verify: show port-security, show port-security interface <int>, show port-security address, show ip dhcp snooping, show ip dhcp snooping binding, show ip arp inspection.

STOP • DEPTH LIMIT

No 802.1X port-based authentication configuration, no MACsec, no storm control.

☐ STUDIED ☐ ANKI ☐ PROVEN

5.8 Compare authentication, authorization, and accounting concepts

CLASS [CONCEPT]

YIELD LOW-MEDIUM

JEREMY DAY 48

- Authentication = who are you. Authorization = what are you allowed to do. Accounting = what did you do (logging, auditing, billing).
- Map each to a scenario. Nearly all questions here are "which of the three is this an example of?"
- TACACS+ vs RADIUS comparison again (see 2.8) — the separation of AAA functions in TACACS+ is what makes it right for device administration.

STOP • DEPTH LIMIT

No ISE, no 802.1X supplicant/authenticator/authentication-server flow in depth.

☐ STUDIED ☐ ANKI ☐ PROVEN

5.9 Describe wireless security protocols (WPA, WPA2, WPA3)

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 57

- The progression and the cipher for each: WEP (broken, RC4), WPA (TKIP), WPA2 (AES-CCMP), WPA3 (AES-GCMP with SAE replacing the PSK handshake).
- Personal (PSK — shared secret, everyone uses the same one) vs Enterprise (802.1X + RADIUS — per-user credentials) for each generation.
- What WPA3 actually fixes: SAE (Dragonfly) defeats offline dictionary attacks against a captured handshake, and adds forward secrecy. Also know Enhanced Open (OWE) for open networks.
- The 802.1X roles: supplicant (client), authenticator (AP/WLC), authentication server (RADIUS).

STOP • DEPTH LIMIT

No four-way-handshake cryptographic breakdown, no EAP method comparison table (PEAP vs EAP-TLS vs EAP-FAST detail).

☐ STUDIED ☐ ANKI ☐ PROVEN

5.10 Configure and verify WLAN within the GUI using WPA2 PSK

CLASS [CONFIG] (GUI)

YIELD MEDIUM

JEREMY DAY 58

Master

- The complete end-to-end flow in a WLC GUI: create the dynamic interface and map it to a VLAN → create the WLAN with SSID and profile name → set the interface on the General tab → on Security → Layer 2, choose WPA2 with AES and PSK → enter the pre-shared key → enable the WLAN → verify a client associates.
- Practise this in Packet Tracer against the WLC device. The GUI there is simplified but the flow matches.
- Verify: client shows connected, receives an address from the correct VLAN's scope, and can ping its gateway.

STOP • DEPTH LIMIT

No WPA2-Enterprise/RADIUS setup, no per-firmware-version menu-path memorisation. WPA2 PSK only — that's literally what the objective says.

☐ STUDIED ☐ ANKI ☐ PROVEN

Domain 6.0 — Automation And Programmability (10%)

Zero configure verbs. Every objective is explain, compare, describe, or recognize. This is pure reading and flashcards — and it is the highest points-per-hour domain on the entire exam, because 10% of the grade requires no lab time at all. Do not skip it, and do not over-engineer it.

6.1 Explain how automation impacts network management

CLASS [CONCEPT]

YIELD LOW-MEDIUM

JEREMY DAY 59

- Traditional per-device CLI management: slow, inconsistent, error-prone, doesn't scale, configuration drift.
- What automation buys: consistency, speed, scale, version control, repeatability, fewer human errors.
- What it costs: upfront skill investment, tooling complexity, and the fact that a mistake now propagates to every device instantly.

STOP • DEPTH LIMIT

No actual scripting. No Python required for CCNA.

☐ STUDIED ☐ ANKI ☐ PROVEN

6.2 Compare traditional networks with controller-based networking

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 62

- Traditional: every device runs its own control plane and data plane; configuration is per-box.
- Controller-based: the control plane is centralised in a controller which has a full network view and programs the devices; the devices keep their data planes.
- Intent-based networking: you declare the outcome, the controller works out the device configuration and continuously verifies it.

STOP • DEPTH LIMIT

No Catalyst Center / DNA Center hands-on, no SD-Access fabric roles in depth.

☐ STUDIED ☐ ANKI ☐ PROVEN

6.3 Describe controller-based, software defined architecture (overlay, underlay, fabric)

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 62

Master

- Underlay = the physical network and the routing that makes it reachable. Overlay = the virtual/tunnelled network built on top (VXLAN, GRE). Fabric = underlay + overlay together as one managed system.
- Control plane vs data plane separation: control plane decides where traffic goes (routing protocols, STP, ARP), data plane actually moves the frames. SDN lifts the control plane out to the controller. Also know the management plane (SSH, SNMP, syslog).
- Northbound API: controller ↔ applications/operators, typically REST/JSON, human-and-app friendly. Southbound API: controller ↔ network devices, typically NETCONF/RESTCONF/OpenFlow. Which direction is which is a guaranteed question — northbound points up toward the user.

STOP • DEPTH LIMIT

No VXLAN header structure, no OpenFlow table pipeline, no LISP.

6.4 Explain AI (generative and predictive) and machine learning in network operations

CLASS [CONCEPT]

YIELD LOW-MEDIUM

JEREMY V1.1 AI/ML VIDEO (NEW IN V1.1)

- Predictive AI forecasts from patterns in existing data (capacity planning, anomaly detection, predicting a failing link). Generative AI produces new content (config drafts, summaries, natural-language queries against network data).
- Supervised (labelled data) vs unsupervised (finds structure itself, good for anomaly detection) vs reinforcement learning.
- AIOps in practice: baselining normal behaviour, flagging deviation, correlating events into a single root cause, recommending remediation.
- Limitations and risks: training data quality, hallucination in generative output, and why a human stays in the loop for network changes.

STOP • DEPTH LIMIT

New v1.1 objective, low depth expected. No model architectures, no training methodology, no vendor product feature lists.

6.5 Describe characteristics of REST-based APIs

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 61

- CRUD ↔ HTTP verb mapping: Create = POST, Read = GET, Update = PUT/PATCH, Delete = DELETE. Know PUT (replace whole resource) vs PATCH (partial update).
- REST constraints: client-server, stateless (each request carries everything it needs — this is the most-tested property), cacheable, uniform interface, layered.
- HTTP status code families: 2xx success (200 OK, 201 Created), 3xx redirect, 4xx client error (400 bad request, 401 unauthorised, 403 forbidden, 404 not found), 5xx server error.
- Authentication types: Basic (base64 user:pass — not encryption), Bearer token, API key, OAuth 2.0.
- Data encoding: JSON (most common), XML, YAML. Request/response headers and body.
- URI anatomy and what a resource path represents.

STOP • DEPTH LIMIT

No writing API client code, no NETCONF YANG model structure, no gRPC.

6.6 Recognize the capabilities of configuration management mechanisms such as Ansible and Terraform

CLASS [CONCEPT]

YIELD LOW-MEDIUM

JEREMY DAY 63 + V1.1 TERRAFORM VIDEO

- Ansible: agentless, push model, uses SSH, written in YAML (playbooks, inventory, modules), procedural/imperative-leaning, focused on configuration management of existing infrastructure.
- Terraform: agentless, uses provider APIs, written in HCL, declarative, maintains state, focused on provisioning infrastructure.
- The comparison axes the exam cares about: agent vs agentless, push vs pull, provisioning vs configuration management, declarative vs procedural, mutable vs immutable infrastructure, and the language/file format each uses.
- Puppet and Chef for contrast: agent-based, pull model, Ruby-based DSL.

STOP · DEPTH LIMIT

Recognize verb — the lowest bar in the blueprint. No playbook authoring, no HCL syntax, no module reference.

☐ STUDIED ☐ ANKI ☐ PROVEN

6.7 Recognize components of JSON-encoded dataCLASS **[CONCEPT]**YIELD **MEDIUM**JEREMY **DAY 60**

- The six data types: string, number, boolean, null, object {}, array [].
- Syntax rules: keys are always double-quoted strings, key-value pairs separated by colons, elements by commas, no trailing comma, no comments.
- The exam skill: given a block of JSON, answer "what is the value of key X" or "how many elements are in array Y" or "which of these is syntactically valid JSON." Practise reading nested structures — an array of objects inside an object is the standard shape.
- Recognise XML and YAML on sight and be able to tell them apart from JSON.

STOP · DEPTH LIMIT

Recognize verb. No parsing code, no JSON Schema, no YAML anchors/aliases.

☐ STUDIED ☐ ANKI ☐ PROVEN

High-Yield Summary — Where The Points Actually Are

If time collapses, protect these in this order. They span roughly two-thirds of the scoring weight and virtually all of the simulation items.

RANK	TOPIC	OBJECTIVES	WHY
1	IPv4 subnetting & VLSM	1.6	Underpins ACLs, OSPF wildcards, NAT pools, DHCP scopes, routing tables. A subnetting error cascades into every other domain.
2	Single-area OSPFv2	3.4	The densest configure-and-verify technology, in the largest domain.
3	VLANs & 802.1Q trunking	2.1, 2.2	The foundation of nearly every switching simulation.
4	ACLs (standard, extended, named)	5.6	Heavily simulated, and unforgiving — wildcard mask or placement errors fail the whole item.
5	Routing table interpretation & forwarding logic	3.1, 3.2	Longest prefix match + AD + metric appears in some form on nearly every attempt.
6	Rapid PVST+ interpretation	2.5	Root election and port roles from a diagram, by hand.
7	IPv4/IPv6 static & floating static routing	3.3	Fast points, easy to lab, easy to get wrong under time pressure.
8	EtherChannel (LACP)	2.4	Mode matrix + consistency troubleshooting.

RANK	TOPIC	OBJECTIVES	WHY
9	IPv6 addressing & types	1.8, 1.9	Compression and EUI-64 are pure speed skills.
10	Layer 2 security (port security, DHCP snooping, DAI)	5.7	Configure-verb security objectives with clear, testable behaviour.
11	SSH device hardening	4.8	A fixed command sequence that appears constantly.
12	NAT/PAT	4.1	The inside-local/inside-global table question is near-guaranteed.
13	Domain 6.0 in full	6.1–6.7	10% of the exam for zero lab time. The best points-per-hour on the blueprint.

MASTER "DO NOT STUDY" LIST Everything below is out of scope for 200-301 v1.1. Some of it appears in Jeremy's course or the OCG for context. Watch once if you like; never lab it, never flashcard it.

- ☐ RIP and EIGRP configuration (not on the v1.1 blueprint — Jeremy Day 25 is context only)
- ☐ BGP in any form
- ☐ OSPF multi-area, LSA types, ABR/ASBR, stub/NSSA areas, virtual links, OSPF authentication
- ☐ OSPFv3 / IPv6 routing protocols
- ☐ VTP configuration (Day 19 is context only)
- ☐ MST, legacy PVST+ timer tuning
- ☐ IPv6 ACLs, time-based ACLs, reflexive ACLs, object groups
- ☐ DHCPv6, SLAAC internals, NAT64, tunnelling mechanisms
- ☐ MQC QoS configuration (class-map / policy-map / service-policy)
- ☐ AAA server configuration (aaa new-model, TACACS+/RADIUS setup)
- ☐ 802.1X configuration, MACsec, storm control
- ☐ IPsec crypto map / IKEv2 configuration, DMVPN
- ☐ Policy-based routing, VRF configuration, IP SLA / object tracking
- ☐ VXLAN, LISP, SD-Access fabric roles, ACI

- ☐ Kubernetes, Docker commands, ESXi administration
 - ☐ Python scripting, NETCONF/YANG models, Ansible playbook authoring
 - ☐ Wireless: 802.11 frame types, RF maths, site surveys, EAP method comparison tables
 - ☐ Any Cisco hardware model numbers or IOS version feature matrices
-

Overview

Part of the CCNA 200-301 v1.1 Study Master Plan — see 00_Index_and_How_To_Use.md for notation/checkbox conventions and 01_Blueprint_Breakdown.md for the objectives referenced by week. Next: 03_Lab_Scenarios.md.

Section 2 — 12-Week Study Calendar

Schedule Parameters

SLOT	TIME	COMPOSITION
Mon–Fri	1.5–2.0 hrs	20 min Anki + 10 min subnetting drills + 60–80 min new content/lab + 5 min verbal recap
Sat or Sun (pick one, keep it fixed)	3.0–4.0 hrs	30 min Anki + 20 min subnetting + 90–120 min lab build + 60 min week review
The other weekend day	0 hrs	Rest. Deliberately. Do not study. Twelve weeks is a long run and a skipped rest day this week becomes a skipped study week in Week 7.
Weekly total	12–15 hrs	

Structural note on phases. The brief specified Phase 1 (Weeks 1–8) for all new content. At 13 hrs/week, Jeremy's ~63 lecture days plus labs does not fit in 8 weeks without shortchanging subnetting and OSPF — the two topics you cannot afford to rush. So this plan carries one deliberate deviation: Domain 6.0 (Automation, 10%, zero CLI, pure flashcard material) moves into Week 9 and runs alongside the start of review. Everything else lands as specified. This is the correct trade: Domain 6 is the easiest content on the blueprint and the least harmed by being studied late. Jeremy day-number caveat. Day numbers below follow his published course ordering. The v1.1 refresh inserted new videos (AI/ML, Terraform, VPN/IPsec) and shifted some numbering. Match by video title, not by number — the titles are stable. Anki protocol, every single day:

- ☐ Reviews cleared to zero before anything else
- ☐ Unsuspend a topic's deck the same day you watch its video — never in advance
- ☐ If the backlog exceeds one day, stop adding new cards until it's clear
- ☐ Never "suspend to catch up" — reduce new cards instead

PHASE 1 — CORE THEORY, DAILY ANKI, GUIDED LABS (Weeks 1–8)

WEEK 1 — Network components, architectures, physical layer

BLUEPRINT 1.1, 1.2, 1.3, 1.4 (INTRO), 2.8 (PARTIAL)

JEREMY DAY 1 (NETWORK DEVICES), DAY 2 (INTERFACES & CABLES), DAY 3 (OSI MODEL & TCP/IP SUITE), DAY 4 (INTRO TO THE CLI), DAY 52 (LAN ARCHITECTURES), DAY 53 (WAN ARCHITECTURES)

- ☐ Mon — Day 1 + Day 2. Install Packet Tracer. Build a two-PC + one-switch topology just to learn the interface.
- ☐ Tue — Day 3. Draw the OSI and TCP/IP stacks side by side from memory, encapsulation top to bottom.
- ☐ Wed — Day 4. CLI navigation drill: user EXEC → privileged EXEC → global config → interface config → line config, and back out. Hostname, banner, save config.
- ☐ Thu — Day 52. Sketch two-tier and three-tier topologies and label what each layer does.
- ☐ Fri — Day 53. Build a comparison table: leased line / MPLS / Metro Ethernet / internet VPN / SD-WAN.
- ☐ Weekend — Review all six days. Start Jeremy's Days 1–4 labs. Begin daily subnetting drills (binary conversion and powers of 2 only this week).
- ☐ Week gate: Explain every device in 1.1 and every topology in 1.2 out loud, notes closed.

WEEK 2 — Ethernet switching, IPv4 addressing, transport, virtualization

BLUEPRINT 1.5, 1.6 (FOUNDATION), 1.7, 1.12, 1.13, 1.4 (FULL)

JEREMY DAYS 5–6 (ETHERNET LAN SWITCHING 1–2), DAYS 7–8 (IPv4 ADDRESSING 1–2), DAY 9 (SWITCH INTERFACES), DAY 10 (IPv4 HEADER), DAY 30 (TCP & UDP), DAY 54 (VIRTUALIZATION PARTS 1–3)

- ☐ Mon — Days 5 & 6. Paper drill: given a frame sequence, fill in each switch's MAC table and mark switched vs flooded.
- ☐ Tue — Days 7 & 8. Classes, masks, RFC 1918 ranges, special addresses.
- ☐ Wed — Day 9 + Day 10. Read show interfaces output; deliberately create a duplex mismatch in PT and observe the counters.
- ☐ Thu — Day 30. Memorise the port-number table today — it pays off in every later domain.
- ☐ Fri — Day 54 (all three parts: hypervisors, containers, VRFs).
- ☐ Weekend — Jeremy's Days 5–10 labs. Subnetting drills move to full network ID / broadcast / host range problems.
- ☐ Week gate: Complete a MAC-table trace exercise with zero errors, and recite all port numbers cold.

WEEK 3 — SUBNETTING WEEK + routing fundamentals [HY]

BLUEPRINT 1.6 [HY], 3.1 [HY], 3.2 [HY], 3.3

JEREMY DAYS 13–15 (SUBNETTING 1–3 INCL. VLSM), DAY 11 (ROUTING FUNDAMENTALS), DAY 11B (STATIC ROUTING), DAY 12 (LIFE OF A PACKET) THIS IS THE SINGLE MOST IMPORTANT WEEK IN THE PLAN. EVERYTHING DOWNSTREAM DEPENDS ON IT.

- ☐ Mon — Day 13. 25 subnetting problems, timed, untimed target.
- ☐ Tue — Day 14. 25 problems. Start timing. Target ≤90 s each.
- ☐ Wed — Day 15 (VLSM). 10 VLSM carve-up exercises on paper.
- ☐ Thu — Day 11 Routing Fundamentals + Day 11b Static Routing. Build a three-router chain in PT and make every PC reach every other PC with static routes only.
- ☐ Fri — Day 12 (Life of a Packet). Narrate a packet's full journey out loud, hop by hop, including ARP and MAC rewrites.
- ☐ Weekend — 50 subnetting problems in one sitting. Then Jeremy's Days 11–15 labs. Then 10 more problems. Target by Sunday night: ≤60 seconds per problem.
- ☐ Week gate: 20 consecutive subnetting problems correct, averaging under 60 seconds. If you miss this gate, repeat Week 3. Do not proceed on a shaky foundation — this is the one week where falling behind is the right call.

WEEK 4 — IPv6 + VLANs

BLUEPRINT 1.8 [HY], 1.9, 1.10, 2.1 [HY], 2.2 [HY]

JEREMY DAYS 31–33 (IPv6 PARTS 1–3), DAYS 16–18 (VLANs PARTS 1–3), DAY 19 (DTP/VTP – CONTEXT ONLY)

- ☐ Mon — Day 31. 30 address compression/expansion reps.
- ☐ Tue — Day 32. Address types by leading bits + 20 EUI-64 conversions by hand, both directions.
- ☐ Wed — Day 33 (IPv6 routing/static). Configure IPv6 addressing and static routes in PT.
- ☐ Thu — Days 16 & 17 (VLANs, access ports, trunking).
- ☐ Fri — Day 18 (VLANs across switches, voice VLAN) + Day 19 (watch VTP once, do not lab it, do not flashcard it).
- ☐ Weekend — Jeremy's IPv6 and VLAN labs. Build a three-switch topology with four VLANs spanning all three switches. Break a trunk deliberately and fix it using show interfaces trunk alone.

- ☐ Week gate: Compress/expand any IPv6 address in under 20 seconds. Build a multi-switch VLAN topology from a blank config in under 15 minutes.

WEEK 5 — Spanning Tree, EtherChannel, discovery protocols • LAB 1

BLUEPRINT 2.3, 2.4 [HY], 2.5 [HY]

JEREMY DAYS 20–21 (STP 1–2), DAY 22 (RSTP), DAY 23 (ETHERCHANNEL), DAY 36 (CDP & LLDP)

- ☐ Mon — Day 20. Bridge ID election by hand from five different diagrams.
- ☐ Tue — Day 21. Port roles and states; port cost values memorised.
- ☐ Wed — Day 22 (RSTP). Map legacy STP states onto RSTP states. Learn the protection features (PortFast, BPDU guard/filter, root guard, loop guard) and exactly what each prevents.
- ☐ Thu — Day 23 (EtherChannel). Build the LACP mode matrix from memory. Lab a bundle that fails due to a config mismatch, then fix it.
- ☐ Fri — Day 36 (CDP/LLDP). Reconstruct a topology diagram from show cdp neighbors detail output alone.
- ☐ Weekend — BUILD LAB 1 (Switched Access Layer). Full build from scratch, no notes on the second attempt.
- ☐ Week gate: Given any STP diagram, mark every port's role and state correctly. Recite the LACP mode matrix.

WEEK 6 — OSPF, FHRP, ACLs • LAB 2

BLUEPRINT 3.4 [HY][HY], 3.5, 5.6 [HY]

JEREMY DAY 24 (DYNAMIC ROUTING), DAY 25 (RIP/EIGRP – CONTEXT ONLY), DAYS 26–28 (OSPF 1–3), DAY 29 (FHRP), DAY 34 (STANDARD ACL), DAY 35 (EXTENDED ACL) THE HEAVIEST TECHNICAL WEEK. OSPF IS THE HIGHEST-VALUE TECHNOLOGY ON THE EXAM.

- ☐ Mon — Day 24 + skim Day 25 (RIP and EIGRP are not on the blueprint — watch for AD context only, 30 min max, then stop).
- ☐ Tue — Day 26 (OSPF fundamentals, neighbour states, RID selection).
- ☐ Wed — Day 27 (DR/BDR election, network types, priority, non-preemption).
- ☐ Thu — Day 28 (cost/metric, passive interfaces, default-information originate). Lab a three-router single-area OSPF domain.
- ☐ Fri — Day 29 (FHRP concepts) + Day 34 (Standard ACLs). 20 wildcard mask conversions.
- ☐ Weekend — Day 35 (Extended ACLs). Then BUILD LAB 2 (STP & Inter-VLAN Routing). Then break an OSPF adjacency five different ways (area mismatch, timer mismatch, subnet mismatch, passive interface, MTU) and diagnose each using show ip ospf neighbor and show ip ospf interface.
- ☐ Week gate: Configure single-area OSPF on three routers from blank configs, no notes, adjacency Full, all routes present, in under 20 minutes.

WEEK 7 — IP Services (Domain 4.0 complete) • LAB 3

BLUEPRINT 4.1–4.9 (ENTIRE DOMAIN)

JEREMY DAY 37 (NTP), DAY 38 (DNS), DAY 39 (DHCP), DAY 40 (SNMP), DAY 41 (SYSLOG), DAY 42 (SSH), DAY 43 (FTP/TFTP), DAYS 44–45 (NAT STATIC & DYNAMIC), DAYS 46–47 (QOS / VOICE VLAN) MANY TOPICS, ALL SHALLOW. KEEP THEM SHALLOW.

- ☐ Mon — Day 37 (NTP) + Day 38 (DNS). Both are short.
- ☐ Tue — Day 39 (DHCP). Configure a router as server, as client, and as relay agent. Lab all three.
- ☐ Wed — Day 40 (SNMP) + Day 41 (Syslog). Memorise severity levels 0–7 today with the mnemonic.
- ☐ Thu — Day 42 (SSH). Type the full hardening sequence from blank, five times, no ?, no Tab.
- ☐ Fri — Day 43 (FTP/TFTP — one pass only) + Day 44 (Static NAT).
- ☐ Weekend — Day 45 (Dynamic NAT/PAT) + Days 46–47 (QoS — explain verb only, no MQC configuration). Then BUILD LAB 3 (IP Connectivity & Dynamic Routing).
- ☐ Week gate: Fill in a NAT translation table (inside local/global, outside local/global) from a scenario with zero errors. Recite syslog severities 0–7.

WEEK 8 — Security + Wireless • LABS 4 & 5

BLUEPRINT 5.1-5.5, 5.7 [HY], 5.8, 5.9, 5.10, 1.11, 2.6, 2.7, 2.9

JEREMY DAY 48 (SECURITY FUNDAMENTALS), DAY 49 (PORT SECURITY), DAY 50 (DHCP SNOOPING), DAY 51 (DAI), V1.1 VPN/IPSEC VIDEO, DAY 55 (WIRELESS FUNDAMENTALS), DAY 56 (WIRELESS ARCHITECTURES), DAY 57 (WIRELESS SECURITY), DAY 58 (WIRELESS CONFIGURATION)

- ☐ Mon — Day 48 (threats/vulnerabilities/exploits/mitigation, AAA, password policy) + the v1.1 VPN/IPsec video.
- ☐ Tue — Day 49 (Port Security). Lab all three violation modes and recover an err-disabled port.
- ☐ Wed — Day 50 (DHCP Snooping) + Day 51 (DAI). Lab them in the correct order — snooping first, then DAI depends on its binding table.
- ☐ Thu — Day 55 (Wireless Fundamentals) + Day 56 (Wireless Architectures, AP modes, CAPWAP).
- ☐ Fri — Day 57 (Wireless Security: WPA/WPA2/WPA3, PSK vs Enterprise).
- ☐ Weekend — Day 58 (Wireless Configuration in the WLC GUI). Then BUILD LAB 4 (FHRP & IP Services) and BUILD LAB 5 (Security & Infrastructure Protection).
- ☐ Week gate: All five labs built at least once. Port security, DHCP snooping, and DAI configured and verified without notes.

PHASE 2 — REVIEW, UNASSISTED BUILDS, WEAK-POINT DRILLDOWN (Weeks 9–10) No new lecture content after Week 9's automation block. From here everything is retrieval practice.

WEEK 9 — Domain 6.0 + full review of Domains 1.0 and 2.0

BLUEPRINT 6.1-6.7, THEN REVIEW 1.0 & 2.0

JEREMY DAY 59 (INTRO TO AUTOMATION), DAY 60 (JSON/XML/YAML), DAY 61 (REST APIS), DAY 62 (SDN), DAY 63 (ANSIBLE/PUPPET/CHEF), V1.1 TERRAFORM VIDEO, V1.1 AI/ML VIDEO

- ☐ Mon — Day 59 + Day 60. JSON reading drills: 10 nested structures, answer "what is the value of X."
- ☐ Tue — Day 61 (REST APIs). CRUD↔verb mapping, status codes, auth types, stateless constraint.
- ☐ Wed — Day 62 (SDN). Northbound vs southbound, overlay vs underlay vs fabric, control vs data plane. Drill until automatic.
- ☐ Thu — Day 63 (Ansible/Puppet/Chef) + Terraform video. Build the comparison table: agent/agentless, push/pull, declarative/procedural, provisioning/config-management.
- ☐ Fri — AI/ML video (predictive vs generative, supervised vs unsupervised, AIOps).
- ☐ Weekend — Full re-read of Section 1 of this document, Domains 1.0 and 2.0. Tick or untick every checkbox honestly. Anything untickable goes on the Weak Point Register below.
- ☐ Week gate: Domain 6.0 fully flashcarded and all three checkboxes ticked for 6.1–6.7.

WEEK 10 — Unassisted lab rebuilds + weak-point attack

BLUEPRINT ALL DOMAINS, REVIEW ONLY

The rule this week: every lab is rebuilt from the task list only. No notes, no Jeremy video open, no ? in the CLI. If you cannot complete a lab this way, that technology is a weak point.

- ☐ Mon — Rebuild Lab 1 unassisted. Time it. Log every command you had to look up.
- ☐ Tue — Rebuild Lab 2 unassisted. Time it. Log lookups.
- ☐ Wed — Rebuild Lab 3 unassisted. Time it. Log lookups.
- ☐ Thu — Rebuild Lab 4 unassisted. Time it. Log lookups.
- ☐ Fri — Rebuild Lab 5 unassisted. Time it. Log lookups.
- ☐ Weekend — Every command you logged this week is now a weak point. Attack them: re-watch only those specific video segments, re-lab only those tasks. Then re-read Section 1 for Domains 3.0, 4.0, 5.0 and tick/untick honestly.
- ☐ Week gate: All five labs rebuilt unassisted. Weak Point Register written and prioritised.

PHASE 3 — PRACTICE EXAMS, SPEED DRILLS, BLUEPRINT AUDIT (Weeks 11–12)

WEEK 11 — First practice exams and remediation

Practice exam protocol, every time

- ☐ Full 120 minutes, timed, no pauses, no notes, no internet, phone in another room
- ☐ Do not check answers mid-exam

- ☐ After scoring, write every missed question's blueprint objective number in the Weak Point Register
 - ☐ Remediate by objective, not by question — if you missed a wildcard-mask question, the problem is 5.6 and 1.6, not that one question
 - ☐ Mon — Practice Exam #1 (Boson ExSim recommended; Jeremy's exams as alternative). Score it. Do not review yet.
 - ☐ Tue — Full review of Exam #1. Read every explanation, including for the questions you got right. Map every miss to a blueprint number.
 - ☐ Wed — Remediation day 1: attack the two weakest domains identified.
 - ☐ Thu — Remediation day 2: attack the next two weakest.
 - ☐ Fri — Subnetting speed session: 40 problems. Target ≤ 30 seconds each. Plus 20 wildcard-mask conversions.
 - ☐ Weekend — Practice Exam #2. Score, review fully, remediate. Then rebuild whichever lab covers your weakest configure-verb domain.
 - ☐ Week gate: Practice exam score trending upward and above 80%. Subnetting at ≤ 30 seconds.
-

WEEK 12 — Final audit and exam

Target: schedule the real exam for the end of this week — Friday or Saturday.

- ☐ Mon — Practice Exam #3. Score, review, remediate.
 - ☐ Tue — Blueprint audit. Walk Section 1 of this document objective by objective, all 53 of them. Any objective without three ticks gets studied today. This is the final sweep.
 - ☐ Wed — Command syntax speed run. On a blank CLI, type from memory: VLAN + trunk config, EtherChannel, OSPF, static and floating static routes, standard and extended ACL, NAT/PAT, DHCP server and relay, SSH hardening, port security. No ?. No Tab.
 - ☐ Thu — Light day. Anki reviews only. 20 subnetting problems. Re-read the high-yield summary table. Confirm exam logistics — ID, test centre or OnVUE system check, arrival time. No new material.
 - ☐ Fri/Sat — EXAM DAY.
 - ☐ Sun — Rest regardless of outcome. Exam-day mechanics:
 - ☐ You cannot go back. Answer each question and commit.
 - ☐ Do simulations when you reach them, but watch the clock — budget roughly 12–15 minutes for a full sim, less for a simlet.
 - ☐ Write out the subnetting cheat sheet (powers of 2, mask-to-prefix table) on the provided whiteboard in the first 60 seconds, before you start.
 - ☐ On any config sim, show running-config first to see what's already there before typing anything.
-

Weak Point Register

Fill this in from Week 9 onward. One row per objective you cannot tick all three boxes for.

BLUEPRINT #	TOPIC	SYMPTOM (WHAT EXACTLY FAILS)	FIX ACTION	RESOLVED
☐				
☐				
☐				
☐				
☐				
☐				
☐				
☐				
☐				
☐				

Slippage Protocol

You are doing this alongside a full-time internship. Weeks will go wrong. When one does, cut in this order — never improvise the order in the moment:

1. Cut first: Domain 4.0 shallow concept topics (FTP/TFTP, SNMP, QoS theory) and Domain 6.0 re-reads. Flashcards will carry these.
2. Cut second: Re-watching videos you have already watched once. Go to your own notes instead.
3. Cut third: The weekend big-block down to 2 hours instead of 4.
4. Never cut: Daily Anki. Daily subnetting drills. Any **CONFIG** lab time.
5. If you lose a full week: push the exam date by a week. Do not compress Weeks 11–12. The practice-exam and remediation cycle is what converts knowledge into a pass.

Overview

Part of the CCNA 200-301 v1.1 Study Master Plan — see 00_Index_and_How_To_Use.md for notation and 02_Study_Calendar.md for when to build each lab.

Section 3 — Packet Tracer Lab Scenarios

Lab Rules

1. Build the topology yourself. Do not download a pre-built file. Placing and cabling the devices is part of the exercise.
2. No CLI answers are given below, by design. Task lists state requirements, not commands. If you cannot produce the command, that is the finding — go back to the video, then return.
3. No ? and no Tab completion on the second and subsequent attempts at any lab.
4. Verify before you move on. A task is not complete until its verification command proves it.
5. Save your work before breaking anything, then break it deliberately. Fault injection is where the real learning happens — each lab ends with a break-and-fix section.
6. Build each lab twice: once with notes (Phase 1), once cold (Week 10). Addressing convention used throughout (adapt freely, but stay consistent):
 - Management/loopbacks: 10.255.255.0/24
 - User VLANs: 192.168.<VLAN>.0/24
 - Router-to-router links: 10.0.<link>.0/30
 - IPv6: 2001:db8:<site>:<subnet>::/64

LAB 1 — SWITCHED ACCESS LAYER

COVERS 2.1, 2.2, 2.3, 2.4, 5.7 (PORT SECURITY), 1.13

BUILD AFTER WEEK 5

REBUILD COLD WEEK 10

TARGET TIME COLD 45 MINUTES

Topology Objective

DEVICE	ROLE	CONNECTIONS
SW1	Access switch	Two links to SW2 (for the EtherChannel), PC1, IP Phone + PC3
SW2	Access switch	Two links to SW1, one link to SW3, PC2
SW3	Access switch	One link to SW2, PC4
PC1, PC2, PC3, PC4	Endpoints	One each to SW1/SW2/SW1/SW3
IP Phone	Voice endpoint	SW1, with PC3 daisy-chained behind it

VLAN plan: VLAN 10 = Sales, VLAN 20 = Engineering, VLAN 30 = Voice, VLAN 99 = Native/Management, VLAN 999 = Parking (unused ports).

Task List

- ☐ T1.1 Set a unique hostname on each switch and disable DNS lookup so typos don't hang the console.
- ☐ T1.2 Create VLANs 10, 20, 30, 99, and 999 on all three switches with descriptive names.
- ☐ T1.3 Assign PC1 to VLAN 10, PC2 to VLAN 20, PC4 to VLAN 10. Configure the ports as static access ports (do not leave them negotiating).
- ☐ T1.4 Configure the SW1 port serving the IP Phone as an access port in VLAN 20 for the daisy-chained PC3, with VLAN 30 carrying voice traffic on the same physical port.
- ☐ T1.5 Bundle the two SW1–SW2 links into a Layer 2 LACP EtherChannel. SW1 must initiate actively; SW2 must respond without initiating. Configure all trunking parameters on the logical interface, not the members.
- ☐ T1.6 Configure the SW2–SW3 link as a static 802.1Q trunk.
- ☐ T1.7 Change the native VLAN on all trunks to VLAN 99. Confirm the change is symmetric on both ends of every trunk.
- ☐ T1.8 Prune the trunks so only VLANs 10, 20, 30, and 99 are permitted. VLAN 999 must not cross any trunk.
- ☐ T1.9 Move every unused switchport into VLAN 999 and administratively shut them down.
- ☐ T1.10 Enable port security on the PC1 and PC2 access ports: maximum two MAC addresses, sticky learning, violation mode restrict.
- ☐ T1.11 Enable port security on the PC4 access port with maximum one MAC address and violation mode shutdown.
- ☐ T1.12 Ensure CDP is running, then additionally enable LLDP globally — remembering that LLDP transmit and receive are separate settings.
- ☐ T1.13 Give each switch a management IP address in VLAN 99 and a default gateway.

Verification Checklist

- ☐ show vlan brief on all three switches — every VLAN exists, every access port is in the right one
- ☐ show interfaces trunk on all three — correct trunk ports, correct native VLAN, correct allowed and active VLAN lists
- ☐ show etherchannel summary — port-channel shows (SU) and both members show (P)
- ☐ show interfaces port-channel <n> — logical interface up
- ☐ show interfaces switchport on the phone port — confirms access VLAN 20 and voice VLAN 30
- ☐ show port-security and show port-security interface <int> — correct max, correct violation mode, sticky addresses learned
- ☐ show port-security address — sticky entries present
- ☐ show running-config interface <int> — sticky MACs written into the config
- ☐ show cdp neighbors detail and show lldp neighbors detail — full neighbour list, correct port mapping
- ☐ show mac address-table — endpoints learned in the correct VLANs
- ☐ ping PC1 → PC4 (same VLAN, across all three switches) — must succeed
- ☐ ping PC1 → PC2 (different VLANs, no routing yet) — must fail. If it succeeds, your VLAN segmentation is broken.

Break and Fix

- ☐ Change the native VLAN on one end of a trunk only. Find and fix it using show interfaces trunk alone.
- ☐ Remove VLAN 10 from SW2 (the transit switch) and observe PC1↔PC4 break. Diagnose before fixing.
- ☐ Set one EtherChannel member port to a different mode or speed. Observe the flag change in show etherchannel summary.
- ☐ Connect a second MAC to the PC4 port and observe the err-disable. Recover the port.

LAB 2 — SPANNING TREE & INTER-VLAN ROUTING

COVERS 2.5, 2.1.C, 3.1, 3.2

BUILD AFTER WEEK 6

REBUILD COLD WEEK 10

TARGET TIME COLD 50 MINUTES

Topology Objective

DEVICE	ROLE	CONNECTIONS
DSW1	Layer 3 distribution switch	Trunks to ASW1, ASW2, and to DSW2
DSW2	Layer 3 distribution switch	Trunks to ASW1, ASW2, and to DSW1
ASW1	Access switch	Trunks to DSW1 and DSW2; PC1
ASW2	Access switch	Trunks to DSW1 and DSW2; PC2
R1	Router (router-on-a-stick)	Single trunk link to ASW1
PC3	Endpoint behind R1's routed segment	—

This is a full triangle-plus-square topology with deliberate physical loops. STP must block something — your job is to control what. VLAN plan: VLAN 10, VLAN 20, VLAN 99 (native/management).

Task List

- ☐ T2.1 Create VLANs 10, 20, and 99 on all four switches. Trunk every switch-to-switch link with native VLAN 99.
- ☐ T2.2 Set the spanning-tree mode to Rapid PVST+ on all four switches.
- ☐ T2.3 Make DSW1 the root bridge for VLAN 10 and the secondary/backup root for VLAN 20. Make DSW2 the root bridge for VLAN 20 and secondary for VLAN 10. Achieve this by manipulating priority — and be able to state the resulting numeric priority value for each VLAN.
- ☐ T2.4 Before verifying, predict on paper which port on each switch will be root, designated, or alternate for each VLAN. Then check whether you were right.
- ☐ T2.5 Enable PortFast on all access ports connecting to PCs. Do not enable it on any trunk.
- ☐ T2.6 Enable BPDU guard on those same access ports.
- ☐ T2.7 Enable root guard on the DSW1 and DSW2 ports facing the access switches, so no access switch can ever become root.
- ☐ T2.8 Configure router-on-a-stick on R1 for VLANs 10 and 20 using subinterfaces, with the correct 802.1Q encapsulation on each and the native VLAN handled correctly. Assign each subinterface the gateway address for its VLAN.
- ☐ T2.9 Verify inter-VLAN routing works through R1. Confirm PC1 can reach PC2.
- ☐ T2.10 Now migrate the routing: enable IP routing on DSW1 and create SVIs for VLANs 10 and 20, taking over the gateway addresses. Remove or shut down R1's role.
- ☐ T2.11 Make sure the SVIs come up — an SVI requires at least one active access port in the VLAN, or the VLAN to be active on a trunk.
- ☐ T2.12 Configure a routed port (or an SVI in a transit VLAN) between DSW1 and DSW2 and add a static or default route so both distribution switches have full reachability.
- ☐ T2.13 Compare the two approaches in writing: when is router-on-a-stick the right answer, and when is an L3 switch?

Verification Checklist

- ☐ show spanning-tree vlan 10 and show spanning-tree vlan 20 on every switch — confirm the intended root for each VLAN
- ☐ show spanning-tree root — root bridge, root cost, and root port per VLAN
- ☐ show spanning-tree summary — mode is Rapid PVST+, PortFast and BPDU guard status
- ☐ show spanning-tree interface <int> detail — port role, state, cost, and link type
- ☐ Confirm the blocked (alternate) port differs between VLAN 10 and VLAN 20. If it doesn't, your priority tuning didn't take effect — that's the whole point of per-VLAN spanning tree.
- ☐ show ip route on DSW1 and DSW2 — connected and local routes for each SVI, plus the transit route
- ☐ show ip interface brief — all SVIs up/up
- ☐ show vlan brief — VLANs active so the SVIs stay up
- ☐ ping PC1 → PC2 (inter-VLAN) — must succeed
- ☐ tracer PC1 → PC2 — confirms which device is actually doing the routing after the migration
- ☐ ping PC1 → each SVI gateway — must succeed

Break and Fix

- ☐ Shut the forwarding uplink on ASW1 and watch RSTP converge. Time it. Compare that convergence to what legacy STP would have taken.
- ☐ Plug a switch into a BPDU-guard-protected PortFast port. Observe the err-disable and recover it.

- ☐ Lower an access switch's priority below the root's and watch root guard put the DSW port into root-inconsistent state. Then fix it.
- ☐ Omit the encapsulation command on one R1 subinterface and diagnose why that VLAN loses its gateway.

LAB 3 — IP CONNECTIVITY & DYNAMIC ROUTING

COVERS 3.1, 3.2, 3.3, 3.4, 1.8, 1.9

BUILD AFTER WEEK 7

REBUILD COLD WEEK 10

TARGET TIME COLD 55 MINUTES

Topology Objective

DEVICE	ROLE	CONNECTIONS
R1	Edge router	R2 (primary link), R3 (backup link), LAN1
R2	Core router	R1, R3, R4, LAN2
R3	Core router	R1, R2, R4
R4	Remote router	R2, R3, LAN3
SW1-SW3	One access switch per LAN	—
PC1-PC3	One endpoint per LAN	—

Four routers in a partial mesh with redundant paths. Every router gets a loopback interface used as its OSPF router ID and management address. Run IPv4 and IPv6 in parallel (dual stack) throughout.

Task List Part A — Addressing and static routing

- ☐ T3.1 Address every router-to-router link from a /30 scheme you design yourself, and every LAN from your /24 scheme. Document the plan before typing anything.
- ☐ T3.2 Create a loopback on each router: R1 10.255.255.1/32, R2 .2, R3 .3, R4 .4.
- ☐ T3.3 Enable IPv6 routing globally and dual-stack every interface with a global unicast address. Use EUI-64 on at least two interfaces and static addressing on the rest, so you have practised both.
- ☐ T3.4 Before any dynamic routing: configure IPv4 static routes on R1 so it reaches LAN3 via R2. Use a fully-specified static route (both exit interface and next-hop).
- ☐ T3.5 Configure a floating static route on R1 for the same destination via R3, with an administrative distance high enough that it only installs when the primary path fails but low enough that it beats nothing else. Test it by shutting the R1–R2 link.
- ☐ T3.6 Configure a default route on R4 pointing toward the core, and confirm the gateway of last resort appears.
- ☐ T3.7 Configure a host route (/32) on R2 to R4's loopback and observe how longest-prefix match selects it over any broader route.
- ☐ T3.8 Configure IPv6 static routes including a default route, and at least one route using a link-local next hop — which requires specifying the exit interface.
- ☐ T3.9 Remove the static routes for LAN-to-LAN reachability (keep the floating static as a deliberate backup) before moving to Part B. Part B — Single-area OSPFv2
- ☐ T3.10 Enable OSPF process 1 on all four routers, area 0 only.
- ☐ T3.11 Set each router's router ID manually to its loopback address. Verify the RID actually changed — remember what is required for a RID change to take effect.
- ☐ T3.12 Advertise all router-to-router links and all LANs into OSPF using wildcard masks (not subnet masks). Also advertise the loopbacks.
- ☐ T3.13 Configure passive interfaces on every LAN-facing interface and on the loopbacks, so no hellos are sent where there are no OSPF neighbours — while keeping those subnets advertised. Use the default-passive approach with explicit exceptions.
- ☐ T3.14 On the R2–R3 link, manipulate OSPF interface priority so that R2 becomes the DR and R3 the BDR on that segment. Then prove the election is non-preemptive by setting R3's priority higher after the adjacency forms and confirming nothing changes until you clear the process.
- ☐ T3.15 On the R1–R2 link, change the OSPF network type to point-to-point and confirm that no DR/BDR is elected on it.
- ☐ T3.16 Change the reference bandwidth on all four routers so gigabit and faster links have distinct costs. Do it on every router — a mismatch causes suboptimal routing.
- ☐ T3.17 Manually set the OSPF cost on one interface to steer traffic down a specific path. Prove the path changed with a traceroute.

- ☐ T3.18 On R1, configure a default route to a simulated internet (a loopback standing in for the ISP) and inject it into OSPF, so R2, R3, and R4 all learn it.

Verification Checklist

- ☐ show ip interface brief and show ipv6 interface brief — every interface up/up with the right address
- ☐ show ip ospf neighbor — every expected adjacency in FULL state, with correct DR/BDR/DROTHER roles
- ☐ show ip ospf interface <int> — network type, cost, hello/dead timers, priority, DR/BDR identity
- ☐ show ip ospf interface brief — quick sweep of all OSPF-enabled interfaces
- ☐ show ip protocols — router ID, passive interfaces listed, networks advertised
- ☐ show ip route ospf — OSPF routes present with AD 110, plausible metrics
- ☐ show ip route — confirm the gateway of last resort is set, and identify every route code present
- ☐ show ip route <specific prefix> — walk through why this route was chosen (longest match → AD → metric)
- ☐ show ipv6 route — IPv6 statics and connected routes present
- ☐ ping PC1 → PC3 and PC2 → PC3 — must succeed
- ☐ ping IPv6 between routers, including a link-local ping specifying the exit interface
- ☐ tracer PC1 → PC3 — path matches what your cost manipulation intended

Break and Fix Break OSPF five separate ways and diagnose each using only show commands before fixing:

- ☐ Mismatched hello/dead timers on one link
- ☐ Mismatched subnet mask on one end of a link
- ☐ A passive interface where a neighbour actually exists
- ☐ Two routers configured with the same router ID
- ☐ An MTU mismatch (watch the adjacency hang in ExStart/Exchange — learn to recognise this signature)

Then: shut the R1–R2 link and confirm the floating static installs. Bring it back and confirm OSPF reclaims the path.

LAB 4 — FIRST-HOP REDUNDANCY & IP SERVICES

COVERS 3.5, 4.2, 4.3, 4.4, 4.5, 4.6, 4.8, 2.8, 5.3

BUILD AFTER WEEK 8

REBUILD COLD WEEK 10

TARGET TIME COLD 45 MINUTES

Topology Objective

DEVICE	ROLE	CONNECTIONS
R1	Gateway A (HSRP)	SW1, and to the core/server segment
R2	Gateway B (HSRP)	SW1, and to the core/server segment
SW1	Access switch	R1, R2, PC1, PC2
SRV1	DHCP / NTP / Syslog / TFTP server	On a different subnet from the clients — the relay is the point
PC1, PC2	DHCP clients	SW1

R1 and R2 both connect to the same client VLAN, providing a redundant default gateway.

Task List Part A — First-hop redundancy

- ☐ T4.1 Address R1 and R2 on the client VLAN with distinct real IPs from the same subnet.
- ☐ T4.2 Configure HSRP group 1 on both routers sharing a virtual IP that will become the clients' default gateway.
- ☐ T4.3 Make R1 the active router by priority, and enable preemption so R1 reclaims the active role after a failure — remembering that HSRP preemption is off by default.
- ☐ T4.4 Identify the virtual MAC address HSRP generates and confirm it matches the expected format for the group number.
- ☐ T4.5 Create a second VLAN and a second HSRP group where R2 is active and R1 standby, so the two routers load-share across VLANs rather than leaving one idle.

Part B — IP services

- ☐ T4.6 Configure R1 as an NTP master with a stratum you choose. Configure R2 and SW1 as NTP clients pointing at R1. Set the correct timezone on all devices.
- ☐ T4.7 Configure a DHCP pool on SRV1's router (or on the server device) for the client VLAN, handing out address, mask, default gateway (the HSRP virtual IP, not a real interface IP — this is the point of the exercise), DNS server, and domain name. Exclude the first ten addresses in the range.
- ☐ T4.8 Configure a DHCP relay agent on the client-VLAN interface of R1 and R2 pointing at the server. Then explain in writing why the relay is required.
- ☐ T4.9 Set PC1 and PC2 to obtain addresses via DHCP. Confirm they receive the correct gateway.
- ☐ T4.10 Configure syslog on all network devices to send to SRV1, at severity level 4 (warning). Then state exactly which severity levels will and will not be sent, and why.
- ☐ T4.11 Configure logging buffered locally as well, and set the buffer size.
- ☐ T4.12 Configure SSH on R1, R2, and SW1: hostname, domain name, RSA keys of at least 1024 bits, a local user with a privilege level and secret, VTY lines set to local login accepting SSH only, version 2 forced, and an exec-timeout.
- ☐ T4.13 Set enable secret on all devices. Apply service password-encryption, then state why it is not real security.
- ☐ T4.14 Apply a VTY access-class so only the management subnet may SSH to the devices.
- ☐ T4.15 Configure a login banner (MOTD) with an appropriate legal warning — and know why "Welcome" is the wrong banner text.

Verification Checklist

- ☐ show standby and show standby brief — group number, state (Active/Standby), priority, preemption enabled, virtual IP, virtual MAC
- ☐ Confirm R1 is active for VLAN A and R2 is active for VLAN B
- ☐ show ntp status — synchronised, correct stratum
- ☐ show ntp associations — the * marks the selected server
- ☐ show clock on every device — times match
- ☐ show ip dhcp binding — leases issued to PC1 and PC2
- ☐ show ip dhcp pool — correct range, excluded addresses, leases in use
- ☐ ipconfig /all on PC1 — address, mask, gateway is the HSRP virtual IP, DNS, and the DHCP server address
- ☐ show logging — destination, severity level, buffered messages
- ☐ show ip ssh — SSH enabled, version 2
- ☐ show ssh / show users — active sessions
- ☐ show running-config | section line vty — transport input, login local, access-class, exec-timeout
- ☐ Actually SSH from one device to another and confirm Telnet is refused
- ☐ ping PC1 → SRV1 — must succeed

Break and Fix

- ☐ Shut R1's client-VLAN interface. Confirm R2 takes over as active and that PC1 keeps pinging with at most a brief interruption. Bring R1 back and confirm preemption returns it to active.
- ☐ Remove the ip helper-address and observe DHCP fail. Diagnose from the client side first, then the router.
- ☐ Set the DHCP pool's default-router to R1's real IP instead of the virtual IP. Fail R1 and observe that clients now lose connectivity — which demonstrates exactly why the virtual IP must be the gateway.
- ☐ Set syslog severity to 7 and observe the volume difference.
- ☐ Attempt SSH from outside the management subnet and confirm the access-class blocks it.

LAB 5 — SECURITY & INFRASTRUCTURE PROTECTION

COVERS 5.6, 5.7, 5.10, 4.1, 1.11, 2.6, 2.7, 2.9

BUILD AFTER WEEK 8

REBUILD COLD WEEK 10

TARGET TIME COLD 55 MINUTES

Topology Objective

DEVICE	ROLE	CONNECTIONS
R1	Edge router — NAT/PAT boundary + ACL enforcement	Inside LAN, and outside link to ISP
ISP	Simulated internet	R1; plus a loopback or server acting as a public host
SW1	Access switch	R1, PC1 (Sales), PC2 (Engineering), SRV1 (internal server), rogue DHCP device, WLC/AP
SRV1	Internal server (HTTP + DNS + DHCP)	SW1
WLC + LAP	Wireless controller and lightweight AP	SW1
Laptop	Wireless client	—
Rogue-DHCP	An extra router/server set up as an unauthorised DHCP server	SW1

VLAN plan: VLAN 10 Sales, VLAN 20 Engineering, VLAN 30 Servers, VLAN 40 Wireless, VLAN 99 Management.

Task List Part A — NAT/PAT

- ☐ T5.1 Address the inside LAN from RFC 1918 space and the R1–ISP link from a public-looking range. Mark R1's inside and outside interfaces correctly for NAT.
- ☐ T5.2 Configure a static NAT mapping so SRV1 is reachable from the internet on a fixed public address.
- ☐ T5.3 Configure dynamic NAT with a pool (a small pool — two or three addresses) plus an ACL identifying the inside source addresses. Generate more simultaneous sessions than the pool can hold and observe what happens.
- ☐ T5.4 Reconfigure to PAT overloading the outside interface address. Confirm many inside hosts now share one public address, distinguished by port.
- ☐ T5.5 From the translation table, fill in a four-column chart by hand: inside local, inside global, outside local, outside global — for one specific session. Part B — ACLs
- ☐ T5.6 Write a standard numbered ACL that blocks the Sales VLAN from reaching SRV1 while permitting everything else. Place it correctly — remember standard ACLs go near the destination — and justify your placement in writing.
- ☐ T5.7 Replace it with a named standard ACL and demonstrate inserting a new permit statement in the middle using sequence numbers, without rebuilding the list.
- ☐ T5.8 Write an extended ACL that: permits Engineering to reach SRV1 on HTTP and HTTPS only; permits ICMP from anywhere to SRV1; denies all other traffic to SRV1; permits everything else. Place it near the source.
- ☐ T5.9 Write a named extended ACL applied outbound on R1's ISP-facing interface that blocks Telnet leaving the network but permits SSH.
- ☐ T5.10 Apply an ACL to R1's VTY lines so only the management VLAN may attempt SSH.
- ☐ T5.11 For each ACL, state in writing what the implicit deny at the end will catch, and prove it with a test that should fail.

Part C — Layer 2 protection

- ☐ T5.12 Enable DHCP snooping globally and on VLANs 10, 20, and 40. Trust only the uplink toward the legitimate DHCP server and the switch-to-switch links. Set a rate limit on untrusted ports.
- ☐ T5.13 Power on the rogue DHCP server on an untrusted port. Confirm clients do not receive addresses from it.
- ☐ T5.14 Enable Dynamic ARP Inspection on the same VLANs, with the same trusted ports. Note the ordering dependency — explain in writing why DAI cannot work without DHCP snooping's binding table first.
- ☐ T5.15 Add an ARP ACL for a statically-addressed host that has no DHCP snooping binding.
- ☐ T5.16 Re-apply port security on all access ports with sticky learning and an appropriate violation mode for each port type.

Part D — Wireless

- ☐ T5.17 Connect the LAP to SW1 and the WLC to SW1. Decide and justify: should the LAP port be access or trunk in this local-mode deployment? Should the WLC port be access or trunk? Configure accordingly.
- ☐ T5.18 Give the WLC a management interface address in VLAN 99 and a dynamic interface mapped to VLAN 40 for wireless clients.
- ☐ T5.19 Confirm the LAP joins the WLC. Identify which CAPWAP tunnels exist and what each carries.
- ☐ T5.20 In the WLC GUI, create a WLAN with your chosen SSID, map it to the VLAN 40 dynamic interface, and secure it with WPA2-Personal using AES and a pre-shared key.
- ☐ T5.21 Set the WLAN's QoS profile appropriately and enable the WLAN.

- ☐ T5.22 Associate the laptop, confirm it receives an address from the VLAN 40 DHCP scope, and confirm it can reach the internet through NAT.
- ☐ T5.23 Walk the WLC GUI tabs (General, Security, QoS, Advanced) and write down what each key setting does — this is objective 2.9's interpret skill.

Verification Checklist

- ☐ show ip nat translations — entries present; for PAT, confirm multiple inside locals share one inside global with different ports
- ☐ show ip nat statistics — hits, misses, pool utilisation, correct inside/outside interface list
- ☐ show ip interface <int> — confirms ip nat inside / ip nat outside and which ACLs are applied in which direction
- ☐ show access-lists — read the match counters. Zero matches on a rule you're testing means the traffic isn't reaching it
- ☐ show running-config | include access-class — VTY ACL applied
- ☐ show ip dhcp snooping — enabled VLANs, trusted interfaces, rate limits
- ☐ show ip dhcp snooping binding — binding table populated
- ☐ show ip arp inspection and show ip arp inspection statistics — enabled VLANs, drops recorded
- ☐ show port-security — all protected ports listed with correct modes
- ☐ show ip route on R1 — default route toward the ISP present
- ☐ WLC GUI: AP shows joined, WLAN shows enabled, client shows associated
- ☐ ping PC2 → SRV1 on permitted ports — must succeed
- ☐ ping/HTTP PC1 → SRV1 — must fail (standard ACL working)
- ☐ Telnet from inside to the ISP — must fail; SSH — must succeed
- ☐ ping PC1 → ISP public host — must succeed (NAT working)
- ☐ ipconfig on the laptop — VLAN 40 address, correct gateway
- ☐ ipconfig on a client with the rogue DHCP server running — must still hold the legitimate address

Break and Fix

- ☐ Reverse the ACL statement order so the deny precedes the permit it was supposed to follow. Observe the first-match failure and fix it.
 - ☐ Apply the extended ACL in the wrong direction on the interface. Diagnose using show ip interface plus match counters.
 - ☐ Remove ip nat inside from R1's LAN interface. Observe NAT silently stop working. This is the single most common real-world NAT fault.
 - ☐ Mark the rogue DHCP server's port as trusted and watch clients get poisoned. Then untrust it.
 - ☐ Enable DAI before DHCP snooping has built any bindings and observe legitimate ARP being dropped.
 - ☐ Put the WLC port in the wrong VLAN and diagnose why the AP never joins.
-

Final Lab Audit

Before booking the exam, all of the following must be true.

- ☐ All five labs built from scratch at least twice
 - ☐ All five labs completed cold — no notes, no ?, no Tab completion
 - ☐ Every break-and-fix scenario diagnosed using show commands before fixing
 - ☐ Lab 1 cold in ≤ 45 min · Lab 2 ≤ 50 min · Lab 3 ≤ 55 min · Lab 4 ≤ 45 min · Lab 5 ≤ 55 min
 - ☐ Every show command in every verification checklist can be typed from memory and its output interpreted
 - ☐ All 53 blueprint objectives in Section 1 have three ticks
 - ☐ Practice exam average above 85% across three separate attempts
 - ☐ Subnetting: any problem solved in ≤ 30 seconds
 - ☐ Weak Point Register fully resolved
-

End of plan. Print Section 1 and tick as you go — the checkboxes are the progress tracker, not a decoration.